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TOGO PUBLIC FINANCE REVIEW (PFR) IN AGRICULTURE AND RURAL CONNECTIVITY SECTORS:

Leveraging Public Investment to
Transform the Rural Economy

Public Disclosure Authorized

September 2024



P179302 - Macroeconomics, Trade and Investment

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ACRONYMS AND ABBREVIATIONS

ARSE	Regulatory Authority of the Electricity Sector
AT2ER	Agence Togolosaie d'Electrification Rurale et des Energies Renouvelables
BCEAO	Central Bank of West African States
CEB	Communaute Electrique du Benin
CEET	Compagnie Énergie Électrique du Togo
DEA	Data Envelopment Analysis
DGB	Budget department
FCFA	Franc CFA
GDP	Gross Domestic Product
ha	hectares
IMF	International Monetary Fund
INSEED	National Institute for Statistics, Economic and Demographic Studies
LOLF	Organic Law on Finance Laws
MEF	Ministry of Economy and Finance
NEP	National Electrification Plan
PFR	Public Finance Review
PIMA	Public investment management assessment
PIP	Public Investment Program
PNIASAN	Programme National d'Investissement Agricole et de Sécurité Alimentaire et Nutritionnelle
PPP	Public-private partnerships
SEZ	Special Economic Zone
SSA	Sub-Saharan Africa
TFP	Total factor productivity
US\$	US dollar
WAEMU	West African Economic and Monetary Union
ZAAP	Planned agricultural development zones

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ABSTRACT

This Public Finance Review (PFR) investigates how improved public investment management could contribute to accelerating rural development and structural transformation. Modernizing agriculture and improving rural connectivity are critical to reduce poverty in Togo as 50 percent of those living under the national poverty line depend on agriculture as their primary source of income. However, low-yielding and rainfed production practices trap most farmers in self-subsistence agriculture, while inadequate rural connectivity infrastructure hinders productivity, market access, and opportunities to attract private investment and improve rural households' welfare. This PFR emphasizes a holistic approach to tackling agriculture productivity and rural poverty in Togo by focusing on both agricultural advancements and infrastructure development. First, it assesses the quality of public investment management in Togo, examining the quality of existing systems for selecting, planning, and executing public projects, and identifying key priorities to improve allocative efficiency, strengthen climate resilience and crowd in private investment to maximize impacts. Second, it investigates the effectiveness of public investment in the agriculture sector and how selected reforms could help boost productivity, limit exposure to climate shocks, and bolster food security. Finally, the PFR proposes an in-depth analysis of public investment in the energy and transport sectors, specifically focusing on strategies to improve rural electrification and rural road connectivity.

The analysis presented in this report shows that the public capital stock remains low in Togo and the quality of infrastructure hampered by a lack of allocative efficiency. Despite rising public investment rates over the last decade, the public capital stock per person is still lower today than in early 2000s, reflecting a lack of resources, weaknesses in public investment management, and exposure to climate shocks. Comparing the level of the capital stock and the quality of public infrastructure in Togo also show that allocative efficiency could be significantly improved. For instance, matching the efficiency of public investment in Cote d'Ivoire could increase infrastructure quality in Togo by 14 percent without additional budget resources, with gains potentially reaching 27 percent by matching the efficiency of Vietnam and 37 percent in the case of Bangladesh. Reforms could go a long way in closing this gap. In fact, if Togo was able to increase its Public Investment Management Assessment (PIMA) score to the best performers in Sub-Saharan Africa, gains in terms of infrastructure quality could range between 14 and 23 percent. Legal and operational changes are proposed to strengthen project selection, budgeting, and implementation, and ensure that climate risks are considered throughout the investment cycle.

Leveraging private sector investment will be essential to accelerate rural development amid tight fiscal constraints. The public sector alone will not be able to meet large investment needs in an environment where fiscal space is limited. The analysis presented in this report suggests that public investment can help mobilize private investment for infrastructure and other development needs when the private sector is convinced of the quality and sustainable use of public resources. In particular, crowding in effects are estimated to be larger when public debt and deficit levels are moderate, when the share of concessional financing is elevated, the tax structure more conducive and the control of corruption more effective. As Togo has room for improvement on most of these fronts, these are important reform avenues to increase the mobilization of private capital to boost infrastructure and public service delivery. Other critical reforms to better leverage the private sector include the need for more transparent and efficient preparation of PPP projects, simplification and greater transparency of public procurement processes, and strengthened regulatory bodies to ensure fair competition.

Improving the allocation and efficiency of public spending in agriculture will help support more productive and climate-resilient practices, and strengthen food security. At present, agriculture in Togo is characterized by low productivity, land intensive and climate-vulnerable farming practices that are unable to guarantee food security for rural or urban populations. Efforts to improve productivity through better access to fertilizers, seeds, irrigation and mechanization have fallen short of targets in recent years, reflecting both insufficient budget allocations and implementation challenges, including failings of certain service providers, inefficiencies in financial execution, and the impact of sequential crises. Financing for

agricultural research, natural resource management and climate change are particularly low, impacting the sector's resilience and capacity to mobilize financing to improve climate adaptation and mitigation. Policy recommendations are identified in this PFR to revitalize agricultural research, improve access to agricultural inputs, strengthen resilience, enhance public sector governance and resource management.

The low level of rural electrification is a major constraint to structural transformation in Togo.

Modernizing agriculture with reliable electricity will unlock transformational changes like irrigation, post-harvest processing, and more extended working hours that can lead to increased agricultural productivity. However, only 25 percent of the rural population has access to electricity in Togo, which is significantly lower than most peers, while urban population have near universal access. Key factors explaining low levels of rural electrification are inadequate levels of public and private sector financing for grid and off grid infrastructures, expensive connection costs and poor service quality. The rural electrification strategy involving the Togolese Agency for Rural Electrification and Renewable Energy (AT2ER), the CIZO project, and Tinga fund have helped improve the situation over time, but progress remains sluggish and governance challenges at the public utility company CEET and AT2ER are major hindering factors, with the poor targeting of subsidized electricity tariffs (benefiting only 24.3 percent of the poor, against 62.7 percent of the non-poor) contributing to divert precious resources from needed investments. Future strategies should entail the improvement of the country's energy supply by reforming the electricity tariff structure, enhancing the performance and regulatory framework of the sector entities, while bolstering the CEET and AT2ER's ability to invest in new capacity in rural area and bolster renewable energy solutions.

Rural road connectivity benefited from important investments over time but remains hindered by poor maintenance and degradation from frequent climate shocks.

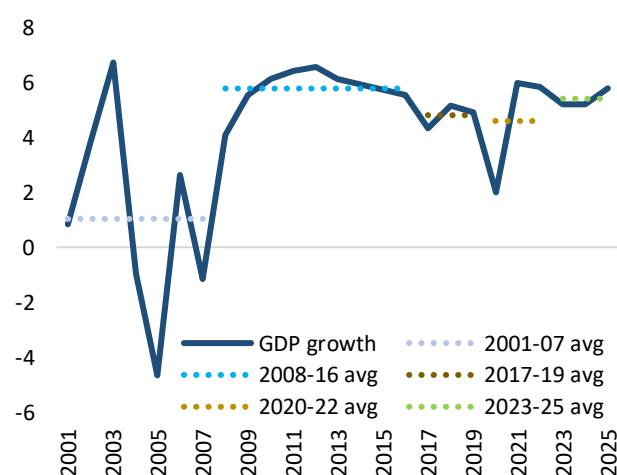
Improving transport connectivity and rural access is crucial to improve access to agricultural inputs thus increasing agricultural productivity, enhance market access, investment opportunities, and food security. The government's efforts to promote connectivity over the past year have been significant, yet rural road access rate in Togo remains relatively low, with 37 percent of the rural population living more than two kilometers from an all-weather road. Many rural roads are unpaved and poorly maintained, making them vulnerable to natural hazards, which in turn hinders access to markets and public services, and discourage private investment in more modern and capital-intensive agriculture practices. While the government's efforts to improve rural road connectivity is commendable, operational inefficiencies, insufficient resources for maintenance and rehabilitation, and limited capacity to build climate-resilient roads are slowing progress. Upgrading the road asset management system, streamlining public procurement processes, fostering private sector engagement through PPPs, prioritizing climate-resilient road network design and construction, and allocating sufficient resources for road maintenance and rehabilitation will be key to sustainably boost rural road connectivity.

COUNTRY CONTEXT AND CHALLENGES

Togo was able to maintain robust growth through challenging times, but structural transformation only made limited headways and fiscal space has been depleted by anti-crisis measures. Since the onset of the COVID-19 pandemic in 2020, the country has faced significant headwinds ranging from global trade disruptions, the fallout from Russia's invasion of Ukraine on energy and food prices, terrorist threats, slowing external demand, and tighter financing conditions. Growth has been resilient in the face of these shocks (Figure 1), mostly thanks to the government's anti-crisis response and a substantial increase in public investment, which reached 9.7 percent of GDP in 2022, up from 3.2 percent in 2019. While this pace is not sustainable given high fiscal deficits and a rising debt burden, investment needs remain large in Togo, including to meet the government's own 2025 Roadmap targets. Despite generally robust growth over the last decade, structural transformation has been hampered by stagnant agricultural productivity and poor connectivity between rural and urban areas, which have slowed the shift towards higher productivity activities and sectors and posed significant hurdles to reducing poverty and mitigating fragility risks. This has contributed to maintaining large disparities in economic opportunities and access to basic services between urban and rural areas and are hindering efforts to reduce poverty and inequality in Togo (Figure 2).

Figure 1. *Togo's growth performance over the past decade has been robust*

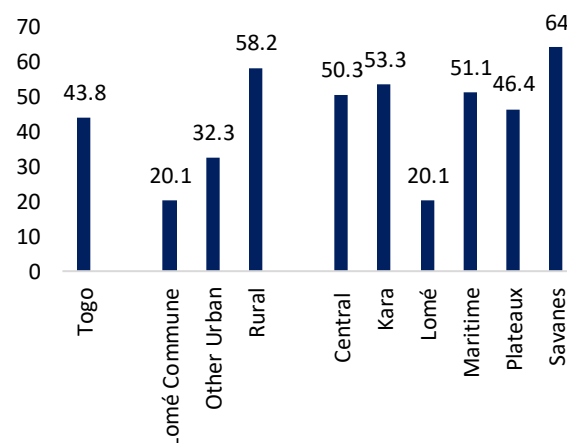
Real GDP growth (percent)



Source: BCEAO, DGEAE, INSEED, and World Bank

Figure 2. *...but poverty remains elevated, particularly in rural areas*

Poverty rate by regions (percent), 2021



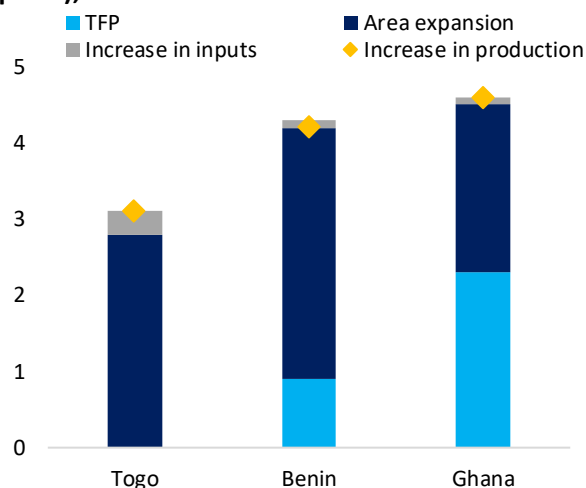
Source: INSEED, World Bank

Modernizing agriculture is one of the most critical building blocks to alleviating extreme poverty and fragility in Togo. Agriculture is the main source of income for two-thirds of rural households and 71 percent of the poor in Togo. Yet the sector faces multiple constraints that are reflected in stagnant productivity and limited opportunities to raise living standards in rural areas. This stems primarily from the low adoption of yield-enhancing practices, with only 37 percent and 8 percent of agricultural households using fertilizer and improved seeds, respectively, and barely 1 percent using irrigation, the lack of adequate infrastructure, with a mere 27 percent of the rural population having access to electricity and 80 percent of rural roads being in poor or average conditions and highly susceptible to climate shocks. As a result, the agriculture sector mostly supplies farmers own consumption and is currently unable to meet the demands of a growing urban population, with only 20 percent of production being commercialized and urban areas relying on imports for food consumption. Stagnant productivity also means that the expansion of agriculture production since the mid-2000s was mainly driven by increases in cropped area rather than efficiency gains, which contributed to deforestation and soil degradation (Figure 3).

Predominance of self-subsistence and rainfed farming practices also mean that variable rainfalls can severely impact the welfare of rural communities.

Figure 3. *Agricultural growth has been driven by cropland expansion, not productivity*

Source of growth in agriculture (percentage point), 2000-2016

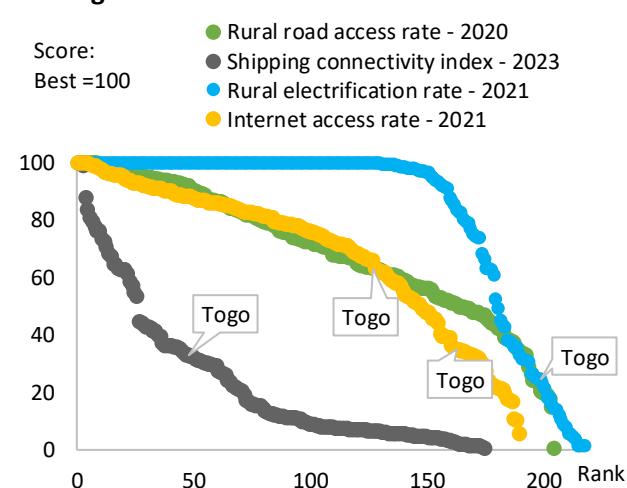


Source: World Bank

Note: TFP = Total Factor Productivity

Figure 4. *Infrastructure access is relatively low, particularly for electricity in rural areas.*

Infrastructure quality scores and global country ranking



Source: World Bank

Better connectivity infrastructures are crucial to untap the potential of the rural economy and alleviate poverty. Studies by Calderón and Servén (2010) show that both the amount and quality of infrastructure have a positive impact on growth and income equality in Sub-Saharan Africa. Improved transportation networks and efficient services can significantly reduce costs for farmers, increasing farm profitability, and allowing them to invest in better seeds, fertilizers, and potentially even electric irrigation pumps, boosting productivity (Dorosh et al. 2012; Tunde and Adeniyi 2012; Iimi 2021). Additionally, better roads and connectivity stabilize food prices, reduce post-harvest losses, and open doors to wider markets, fostering agribusiness development. Expanding the electricity grid further empowers farmers by allowing for mechanization, leading to a more productive and resilient agriculture sector. However, evidence also highlights the importance of tailored approaches, as the impact of infrastructure improvements can vary greatly depending on local conditions and the type of agricultural systems in place (Ashkenazy et al. 2018). These findings underscore the need for integrated and context-specific infrastructure development strategies to maximize the benefits for agricultural productivity.

Underdeveloped and poorly maintained transport and energy connectivity continue to limit development opportunities. Despite recent progress, access to energy and transport connectivity underperform regional peers, with about 75 percent of the rural population lacking electricity access (Figure 4). The 2018 National Electrification Strategy planned grid extensions, the expansion of mini-grid connections, and solar home systems but progress has been slow, partly due to the limited financial and operational capacity of the national electricity company Compagnie Énergie Électrique du Togo (CEET) and the rural electrification and renewable energy agency Agence Togolaise d'Électrification Rurale et des Énergies Renouvelables (AT2ER) to support necessary investments. Regarding the transport sector, significant efforts were made towards helping Togo becoming a regional trade and logistics hub, with major investments to promote connectivity with hinterland countries, as well as special economic zones (SEZs) and zones serving strategic hubs. However, beyond the N1 connecting the Lomé logistics hub (the port, airport, and logistics facilities) and the border with Burkina Faso at Cinkassé, the N2 between Aflao (border with Ghana) and Hillacondji (border with Benin), and the Great Lomé bypass road between the

Port of Lomé and Adétikopé, other parts of the road network are generally in poor or average conditions. Substantial improvements are needed in road maintenance and axle load control, particularly in the secondary network and feeder roads.

The Togo 2025 Roadmap sets out ambitious development goals that translate into equally ambitious investment plans in the coming years. The program outlines a portfolio of 36 projects and 6 priority reforms for the government that span across all sectors of the economy, prioritizing public administration, social harmony, and infrastructure modernization. To drive inclusive economic growth and create jobs, the plan focuses on boosting agriculture (yields, incomes, and food security), expanding electricity access, and modernizing transportation (targeting 80 percent of roads in good condition). The roadmap estimates investments between CFA 2,800–3,400 billion (US\$4.9–6.0 billion), with about 52 percent contribution expected from the private sector. The total costs of the needs in the agricultural sector are evaluated at around 3.4 percent of GDP between 2023 and 2025, with around 32 percent to be financed through domestic funding. Transportation projects are estimated to cost about 8.2 percent of GDP, with 90 percent financed domestically, while projects in the energy sector will require about 3.0 percent of GDP, with funding mainly coming from external sources.

Meeting the infrastructure and development needs, notably amid tight fiscal constraints, will require not only efficient domestic resource mobilization, but also innovative solutions to leverage private sector investment. The public sector alone will not be able to meet large investment needs to modernize agriculture and connectivity infrastructures in an environment where fiscal space is limited and the government capacity to manage projects is constrained. As the government is planning to bring the budget deficit from 6.6 percent of GDP in 2023 to 3 percent by 2025 through ambitious fiscal consolidation measures, achieving the government's 2025 roadmap objectives will only be possible through greater selectivity and efficiency in the management of public investments combined with the mobilization of private capital for key infrastructure and development projects. In this context, Public-Private Partnership (PPP) could be critical for sharing risks and efficiently mobilizing private capital. Well managed PPPs can facilitate innovation, cost efficiency, and improved service quality. But to attract quality investors and stimulate more generally the crowding in of private investment, public sector governance, fiscal sustainability and the quality of the business environment play a critical role. In fact, poor public financial management and institutional capacity can distort the allocation and quality of public investment, potentially leading to infrastructure deficiencies that undermine private sector participation in those projects and in the broader economy.

This Public Finance Review (PFR) brings new analysis and key recommendations to bolster rural development and accelerate structural transformation through better public investment management. More specifically, the first section presents an analysis of the quality of public investment management systems and investigates ways of improving selection, planning and execution, mobilizing private investment and ensuring that future infrastructures are climate resilient. Section 2 assesses the allocation, efficiency, and effectiveness of public investment in the agriculture sector to bolster productivity and job creation while adapting to climate change and supporting food security. Section 3 proposes a deep dive on the energy sector with a view to boosting rural electrification and renewable energy generation while section 4 focuses on the transport sector, identifying key recommendations to increase rural road connectivity. Section 5 concludes with a list of priority reforms.

PUBLIC INVESTMENT MANAGEMENT AND STRUCTURAL TRANSFORMATION

1. Introduction

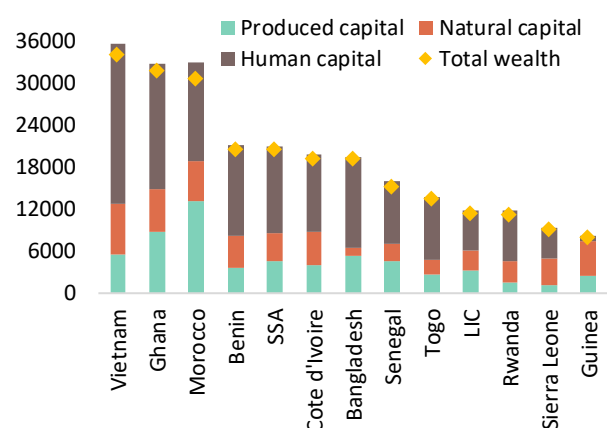
Progress towards a more productive and resilient economy would require faster structural transformation. Sustained economic growth and poverty reduction, as seen among some aspirational peers like Vietnam or Bangladesh, requires shifting resources from low-productivity and labour-intensive activities to higher productivity and more skill-intensive ones. This process of structural transformation can result in significant welfare gains for the population as well as a more productive, diversified and resilient economy. While Togo has made some strides, notably with the expansion of port facilities in Lomé and the development of industrial and agro-business zones, persistent challenges hinder the pace of this transformation, including limitations in the development of a more modern and climate resilient agricultural sector and the lack of quality infrastructure and public services in rural areas. Low productivity in agriculture reflects limited access to yield-enhancing inputs (seeds, fertilizers, phytosanitary products) and irrigation, low levels of mechanization, a lack of agroecological knowledge, and limited access to market and finance. Other factors preventing faster structural transformation are a high degree of concentration of private sector activity, widespread informality, lopsided urbanization, a lack of human capital, and constrained access to finance and know-how. The slow pace of structural transformation poses significant hurdles to mitigating poverty and fragility risks in Togo, highlighting the need to better leverage limited public resources to lift those constraints.

The level and effectiveness of public investment have not been sufficient to ensure a rapid increase in productive capital in Togo. Productive capital is still relatively low in Togo and contributes less to GDP growth than among aspirational peers, while human and natural capital are generally more comparable (Figures 5). This is reflected in lower average rates of public and private investment despite recent improvements (Figure 6). Structural transformation in Togo is also slowed by low levels of agricultural productivity and poor connectivity between rural and urban areas (Azolibe and Okonkwo 2020; Malah Kuete and Asongu 2023), which is exacerbated by the vulnerability of rural areas to climatic hazards.

The government set ambitious goals to accelerate rural development and structural transformation, but investment allocation and execution rates have been volatile, limiting their impact. Public investment over the last decade was primarily focused on economic transformation, with the transport sector receiving the most funding, followed by energy, and agriculture. However, budget commitments and execution in those sectors have fluctuated sharply between periods of expansionary fiscal policy and periods of fiscal consolidation, hampering steady improvements in infrastructure quality and agricultural productivity (Figure 7). As the government enters a new period of fiscal consolidation, its investment program has been scaled back and difficult arbitrages will need to be made (Figure 8). Delivering on ambitious development goals in a tight fiscal framework will require a heightened focus on selectivity, effective project management and execution, private sector engagement, and innovative financing solutions.

Figure 5: *Produced capital in Togo is lagging structural and aspirational peers*

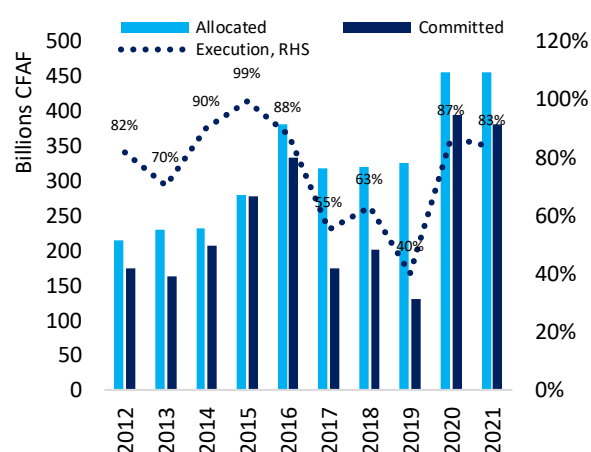
Wealth per capita (USD per capita), 2018



Source: The Changing Wealth of Nations (2022)

Figure 5. *Public investments have suffered from fluctuating commitments and execution rates*

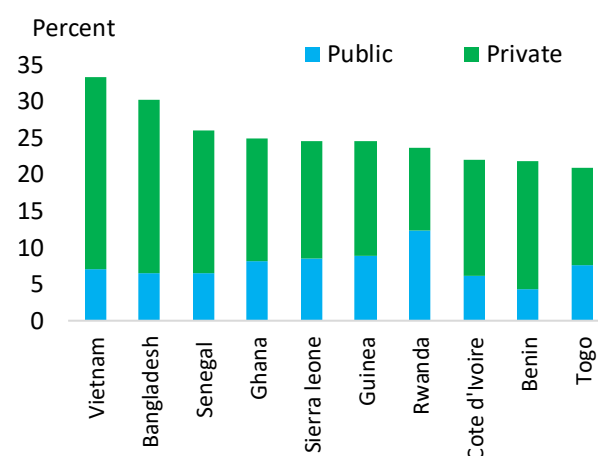
Public investment allocation and execution



Source: BCEAO, BOOST

Figure 6: *Despite an upward trend, investment rates in Togo still trail peers*

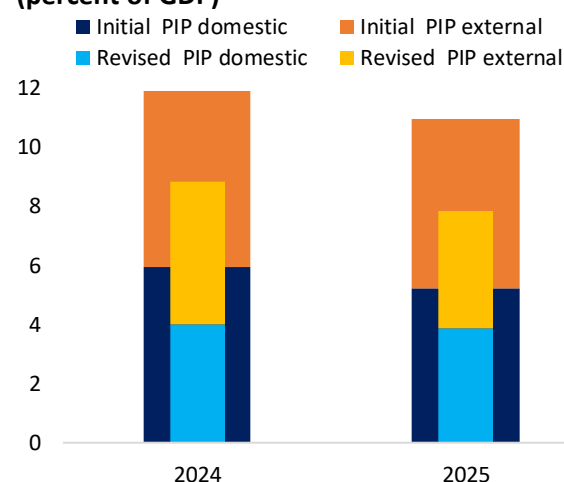
Public and private investment (percentage of GDP), 2010-2022



Source: World Bank

Figure 6. *Ongoing fiscal consolidation efforts are impacting public investment plans*

Initial vs revised investment plan for 2024-25 (percent of GDP)



Source: Ministère de l'Economie et des Finances
Note: PIP = Public Investment Plan

Investment in climate-resilient rural roads could also generate benefits that far exceed the initial cost.

The incremental cost of building resilient roads is generally low compared with long term benefits. For example, the additional cost of making exposed new transport assets more resistant to floods and landslides is estimated at just 0.6 percent on average and up to 5 percent in the most expensive cases, but disaster damage could be reduced by 50 percent, far exceeding the initial cost (Espinete et al. 2018). Information on the risks associated with infrastructure projects is essential to support investment decisions and reduce the incremental cost of building more resilient infrastructure. For instance, the World Bank has developed a Resilience Rating System (RRS), which can be used to assist in project financing and implementation decisions, or to monitor how climate change is taken into account in the decision-making process. This assessment characterizes the extent to which projects explicitly contribute to the resilience of beneficiaries, communities, asset networks or even countries. This resilience rating system can help improve the efficiency and transparency of public investment management systems and

give visibility to the use of hazard and climate change data for infrastructure operations or regular updates of building codes and infrastructure sector regulations.

2. Assessing the allocative efficiency of public investment

Public investment can be an important catalyst for growth and poverty reduction if it is correctly selected and implemented. Empirical research underlines that the positive relationship between public investment, infrastructure quality and growth is closely linked to the selectivity and efficiency of investment projects (Calderón and Servén 2004; Gupta et al. 2014; Calderón, Cantu, and Chuhan-Pole 2018). For instance, a one percent of GDP increase in public investment tends to raise output by just 0.3 percent in low efficiency countries, compared to 0.6 percent for high-efficiency ones (Calderón, Cantu, and Chuhan-Pole 2018)¹. Strengthening public investment management (PIM) could therefore yield significant development gains, particularly in an environment where domestic resource mobilization is weak, and capacity is constrained.

Weaknesses in budgeting, asset management, and transparency hinder the country's ability to fully leverage public investment for structural transformation. Recent assessments of public investment management in Togo shows good performance on fiscal targets, strategy, and planning, and progress made on project selectivity, but inefficiencies remain in institutional and legal framework, budgeting, asset monitoring, and transparency. Togo does not yet have general regulations governing the entire public investment management cycle. While guidelines require sectoral ministries to estimate recurrent costs like upkeep and maintenance, in practice, these costs are often omitted. The allocation of funds to individual projects also lacks consistency and transparency, with commitment and payment appropriations not fully applied as stipulated in the Organic Law on Public Finance Laws (LOLF). Budget envelopes at the program level are global and provisional, leading to the cancellation of unspent investment credits at the fiscal year's end. Project financing faces rationing due to severe cash constraints and inefficient cash management, resulting in implementation delays, payment delays, and arrears. And while project implementation is monitored, including by the Ministry of Development Planning and Cooperation, reports are not published regularly. Furthermore, ex-post evaluation practices are minimal and there's no comprehensive inventory or data on assets' vulnerability to climate risks, indicating a nascent stage in non-financial asset management.

More explicit climate scoring of investment projects is a key priority. Investment in more resilient infrastructure generates benefits that can far exceed the initial cost. For example, in the electricity sector, if exposed assets were made more resilient to natural hazards, there would be an increase in construction costs of around 3 percent on average, and 6 percent in the costliest cases, but the risk of damage would be reduced by a factor of two to three for new infrastructure. However, the current public investment management framework in Togo does not adequately consider climate risks, leading to projects that are vulnerable to climate shocks and could hence fail to deliver expected long-term benefits. By taking better account of climate risks throughout the project lifecycle – from appraisal (especially for domestically financed projects) to selection, planning, design, and implementation – public investments could become a critical driver of climate resilience and transition to a low-carbon economy. Key priorities include better integrating climate considerations into national and sectoral development plans and strategies (notably during initial appraisal), raising awareness among government officials, policymakers, and the public, about the risks of climate change, developing the capacity to assess risks to infrastructures and establishing monitoring systems to track the impact of climate change on investment projects. In this

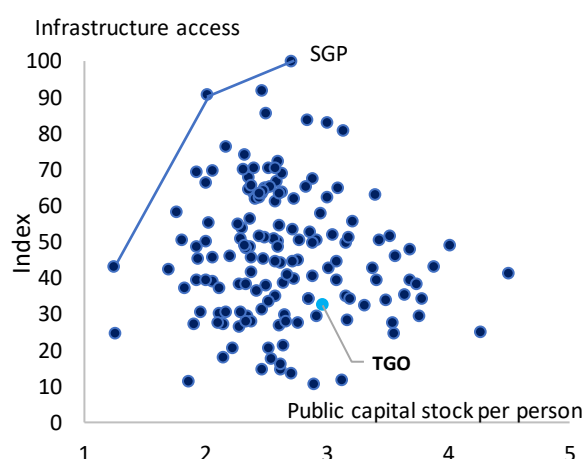
¹ This evidence is consistent with other empirical studies of the relationship between investment and growth such as IMF (2014); and Gupta et al. (2014).

regard, a Resilience Rating System could be operationalized to assist in project financing and implementation decisions and support updates of building codes and infrastructure sector regulations.

Measures of public investment efficiency confirm that Togo could significantly improve public service quality with existing resources. Public investment efficiency² is considered here as the relationship between the level of public investment and the quality of infrastructure services. It is calculated as the distance to an estimated efficiency frontier, which is defined by the set of countries with the best infrastructure quality for a given level of public capital stock per person. Analysis based on the non-parametric Data Envelopment Analysis (DEA)³ show that Togo has an efficiency score of 0.32 using a composite index of infrastructure quality including rural electricity access, rural road access, internet access and shipping access. This score means that Togo could potentially improve infrastructure services by up to 68 percent without increasing the stock of public capital per person if it is able to match the allocative efficiency of Singapore, the best performer in a sample of 153 countries included in the analysis (Figure 9). Using more appropriate comparators, Togo could improve infrastructure quality by 14 percent if it could match the efficiency score of Cote d'Ivoire, 27 percent by matching the efficiency score of Vietnam and 37 percent in the case of Bangladesh (Figure 10). Alternative infrastructure quality indicators were used for robustness checks, most yielding similar results.⁴

Figure 7. Access to infrastructure is well below what Togo's public capital stock could deliver

Infrastructure access and public capital stock, 2021

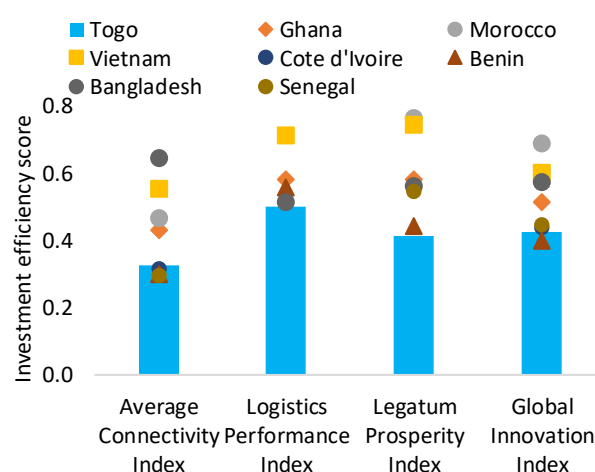


Source: World Bank

Note: This graph plots per person public capital stock on the horizontal axis and realized investment output on the vertical axis. The efficient frontier curve shows which countries have the best combination of the two factors. Countries below the efficient frontier are less efficient because they offer lower output for the same level of input.

Figure 8. Alternative infrastructure quality indicators confirm low investment efficiency

Public investment efficiency scores, 2021



Source: World Bank

Note: This investment efficiency score helps assess how well an investment utilizes resources to generate returns. A higher value indicates better effectiveness of resource allocation. The Average Connectivity Index is the simple average of the rates of rural electricity access (2021), rural road access (2020), internet access (2021) and shipping access (2023).

² The investment efficiency measure is computed using average values of 2015-2019 investment and capital stock data as input, and the most recent value of the infrastructure connectivity measures as output.

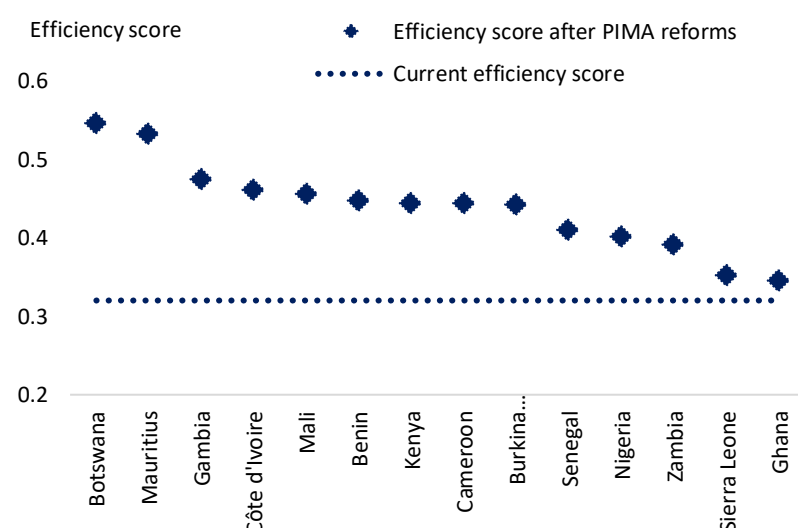
³ The non-parametric Data Envelope Analysis was selected to estimate efficiency scores as it imposes less functional restrictions and allows for multiple input and output variables. The lack of assumptions about statistical distributions means that there are no estimates or significance tests for efficiency scores and can result in sensitivity to the specification of inputs and outputs.

⁴ Alternative measures include the Infrastructure and Market Access score of the Legatum Prosperity Index (2023), the quality of infrastructure in the Global Innovation Index (2023) and the quality of trade and transport related infrastructure (e.g., ports, roads, railroads and information technology) in the Global Logistics Performance index (2023). These are survey-based indicators, while the primary indicator used in this analysis measures actual infrastructure access, with a focus on rural areas.

Public investment management reforms could be associated with significant efficiency gains. The potential benefits of such reforms are estimated using cross-country regressions of 34 Sub-Saharan African countries using DEA efficiency scores as the dependent variable and Public Investment Management Assessment (PIMA) scores⁵ as explanatory variables. Results confirm a positive and highly significant relationship⁶ between PIMA scores and public investment efficiency across Sub-Saharan African countries, illustrating that reforms could have meaningful impacts on infrastructure quality, even without increased budget allocations. In fact, if Togo was able to increase its PIMA effectiveness score to match best performers in the sample (Botswana, the Gambia, and Cote d'Ivoire), gains in terms of infrastructure quality could range between 14 and 23 percent at unchanged public investment levels (Figure 11). Sound and effective institutions are key for better efficiency. Regression results confirm a positive and significant relationship between governance indicators and public investment efficiency scores. Notably, improved government effectiveness tends to ensure that investments are effectively and strategically allocated, while ensuring efficient planning, execution, and monitoring of the projects. Hence, improving the quality of institutions and state-owned enterprises in Togo (e.g., ATE2R, CEET, SAFER as discussed in this report) is also important to enhance the country's ability to deliver on its ambitious investment goals.

Figure 9. *Expected efficiency gains are large if Togo raises its PIMA score to best performing Sub-Saharan Africa peers.*

Estimated efficiency gains from matching PIMA scores of selected SSA countries



Source: World Bank

Note: Based on most recent official PIMA scores, which in the case of Togo date back from 2016

Priority reforms could help boost allocative efficiency, strengthen implementation, monitoring and ensure climate-resilient investments projects. On the legal front, Togo could establish an effective regulatory and institutional framework clarifying responsibilities of the various players and their interactions, as well as the procedures and standards to be applied throughout the investment management cycle. On an operational level, the manual of procedures for selecting, prioritizing, and programming public investment projects could be reinforced, ensuring greater coherence of the national

⁵ The analysis uses Togo's 2016 PIMA scores (last official data) and could therefore overstate the gains of reforms given the progress achieved since then. An updated PIMA assessment scheduled for 2023 is still pending.

⁶ The specified OLS regression highlights only a correlation between public efficiency and the institutions governing public investment management systems, and not a causality.

planning and the medium-term budgetary framework as well as of climate strategy objectives. This involves aligning investments with strategic goals, integrating recurring costs into budgeting, implementing projects on time and within budget, systematizing the production and publication of feasibility studies, better coordinating procurement, commitment, and cash flow plans, publishing periodic reports on project implementation, improving the quality of public asset accounting, developing rules and procedures to enable a systematic assessment of climate adaptation and mitigation dimensions in project planning.

The integration of the spatial dimension in the planning phase of public investments is essential to ensure a balanced and efficient distribution of resources and infrastructure throughout the territory.

Although Togo's 2019 manual of procedures for the selection, prioritization and programming of public investment projects stresses the importance of taking geographical specificities and regional needs into account when evaluating and prioritizing projects, there are no rules guiding the integration of the spatial dimension into the investment planning phase. International best practice recommends that national and sectoral strategies should be guided by a clear vision and sectoral plans that take account of regional disparities and local infrastructure needs. In addition, these various strategies must also be based on robust ex-ante analyses that integrate geographical criteria, to assess the spatial impact of the projects that will be proposed for the pre-budget selection phase.

3. Mobilizing the private sector

Strengthening public service quality in a fiscally constrained environment necessitates mobilizing private sector investment. Private sector involvement in infrastructure development is crucial to bridge the investment gap, reduce fiscal burdens, improve risk management, and boost efficiency and innovation in project execution, ultimately leading to better infrastructure service delivery. Additionally, the active participation of the private sector would foster innovation and competition and improve benefits for end-users. However, unlocking this potential largely depend on allocative efficiency as well as broader governance and business climate considerations (Cavallo and Daude 2011; Bahal, Raissi, and Tulin 2018; Matvejevs and Tkacevs 2022; Chinzara, Dessus, and Dreyhaupt 2023). According to a threshold analysis of the relationship between public and private investment, crowding in effects tend to be larger when initial public investments are low, public debt and deficit levels are moderate, the share of concessional financing is larger, the tax structure and control of corruption efficient (Table 1). In Togo, most of these factors seems to deter private investment at the current juncture, particularly public debt levels, insufficient reliance on concessional financing, the tax structure and the control of corruption, indicating the potential for growth-enhancing reforms in these areas (Figure 12).

Other reforms needed to better leverage private sector investment are to ensure a more transparent and efficient preparation of PPP projects. This includes reviewing the organization of responsibilities within government and agencies regarding the conduct of PPPs, simplify public procurement processes to ensure more competitive bidding, and strengthen regulatory bodies to ensure fair competition. Estimates suggest that four years of continuous improvements in reforms related to the regulatory framework would generate an additional 0.8 percent of GDP above the baseline of no reforms of Private Participation in Infrastructure (Chinzara, Dessus, and Dreyhaupt 2023). In addition, improving the transparency and efficiency of preparation of PPP projects, reassessing the distribution of responsibilities within government and agencies regarding the conduct of PPPs, streamlining and clarifying public procurement processes to promote competitive bidding, and strengthening regulatory bodies to ensure a fair and competitive environment are commendable in order to leverage private investment. The 2023 PEFA evaluation highlighted that, although progress had been made in improving transparency in public procurement,

several challenges remained, particularly the need to disseminate information more widely to the public, to guarantee optimum transparency and prevent fraudulent practices.

Table 1. Thresholds analysis of the relationship between public and private investment

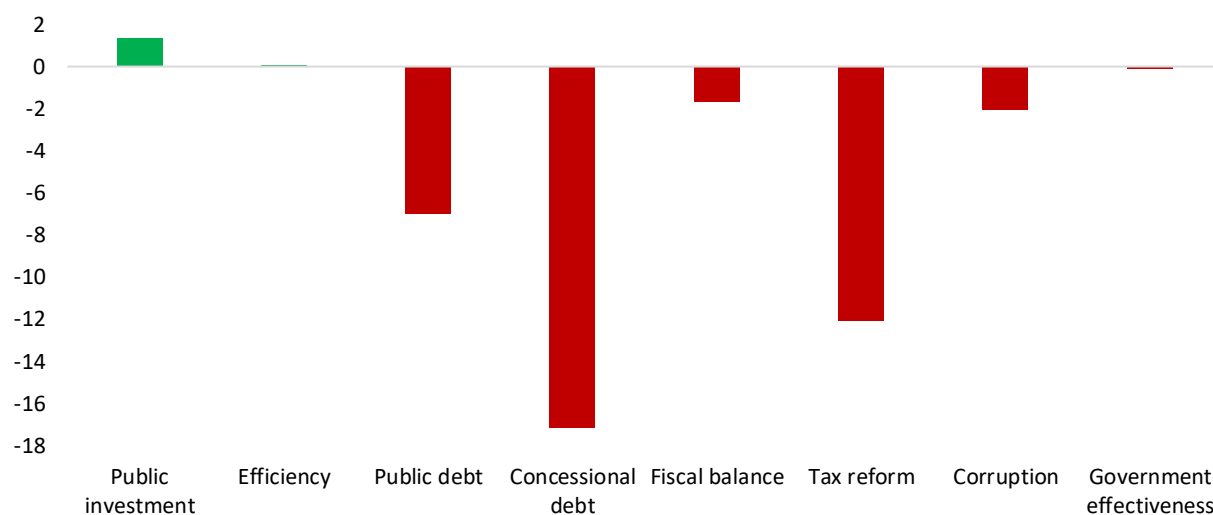
Threshold variables	Threshold value	90% confidence interval		Crowding in	
		Lower	Upper	Below threshold value	Above threshold value
Public investment	5.413	1.921	6.005	0.866**	0.198
Public investment efficiency	1.187	1.109	1.29	0.176***	0.060*
Public debt	46.74	35.78	50.11	0.123**	0.044
Concessional debt	34.04	31.81	34.99	0.371***	0.999***
Fiscal balance	-1.063	-5.262	0.98	0.043	0.711***
Tax reform	40.28	39.79	40.93	-1.125***	0.362**
Control of corruption	1.193	1.169	1.239	0.078	0.339***
Government effectiveness	-1.121	-1.25	1.024	0.008	0.142*

Source: World Bank

Note: *** p<0.01, ** p<0.05, * p<0.1. The threshold value is estimated based on Kremer, Bick, and Nautz's (2013) dynamic panel threshold methodology on a panel of 114 developing economies. The crowding in column denotes the estimated impact of public investment on private investment. The threshold parameter's confidence interval estimation is based on Hansen's (1999) method.

Figure 10. *Most promising avenues to crowd in private investment in Togo is to cap public debt, increase the share of concessional financing, reform taxation and improve the control of corruption*

Gap from estimated threshold values for crowding in effects in Togo



Source: World Bank

Note: The gap is computed as the difference between the average for Togo and the estimated threshold. Green bars suggest that Togo is in the crowding-in zone, while red bars indicate a crowding-out effect or that public investment has no statistically significant effect on private investment.

4. Key policy options

To reinforce public investment management, ensure better alignment with strategic objectives and leverage private sector investment, the following recommendations are identified:

- First, plan sustainable levels of investment across the public sector by strengthening the coherence and traceability of the chain between strategic planning, sectoral planning, and the medium-term budget framework; systematically costing strategies and ensuring that they are accompanied by concrete targets; and ensuring the production and publication of feasibility studies for all investment projects.
- Second, integrate recurring costs into public investment budgeting, particularly in terms of upkeep and maintenance; develop cost standards and guidelines; and guarantee resources for the operation and upkeep of assets.
- Third, develop and implement an integrated geographic information system for public investment planning. This system aims to visualize and analyze spatial data (demographic distribution through population density by region; level of access to basic infrastructure; priority economic zones; projects underway; areas of vulnerability to the impact of climate change), thus facilitating informed decision-making on the location of projects according to regional needs and national priorities. The use of such a system will strengthen the Government's ability to allocate resources more equitably and monitor the spatial impact of investments.
- Fourth, implement projects on time and on budget by improving continuity in the allocation of resources to investment projects; by fully implementing the management of commitment and payment credits (AE-CP) and the credit carry-forward mechanism, and by strengthening coordination between procurement, commitment, and cash flow plans.
- Fifth, reinforce climate resilience in public investment management, by developing a coherent and costed climate change strategy, develop rules and procedures in the methodology of environmental and social impact studies to include criteria that take account exposure to climate shocks and climate mitigation objectives.
- Finally, better leverage the private sector by improving the transparency and efficient preparation of PPP projects; by simplifying public procurement to ensure more competitive bidding, by strengthening regulatory bodies to ensure fair competition, and by reinforcing land and property rights as well as business dispute resolution mechanisms.

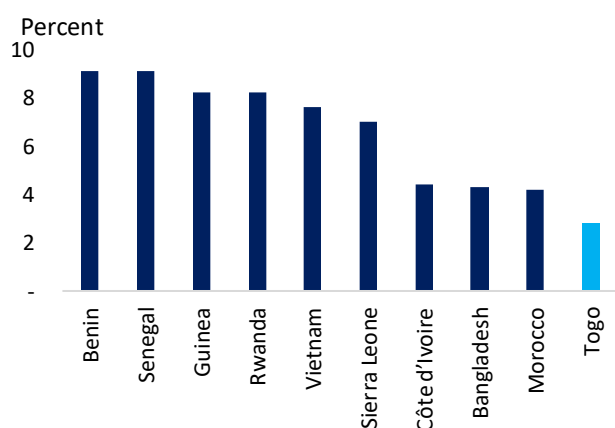
LEVERAGING PUBLIC INVESTMENT TO MODERNIZE THE AGRICULTURE SECTOR

1. Introduction

Sustained economic growth and poverty reduction would require a more modern and resilient agriculture sector. Agriculture is the backbone of Togo's economy but its capacity to drive structural transformation and guarantee food security for its population has so far been limited. Agriculture employs 42.7 percent of the active age population in Togo and up to 70 percent of the rural population, yet the agricultural landscape faces multiple constraints that hinder investments and the adoption of more productive and climate resilient practices, which limit opportunities to raise living standards in rural areas, support the development of industrial and service sectors and reduce factors of fragility and instability. In fact, the sector's diminishing share in overall GDP and slower productivity gains than many peers (Figure 13) illustrate the sector's unrealized potential. Insufficient energy and transport infrastructure impede, mechanization and access to input, limiting productivity and commercialization potential (Calderón, Cantu, and Chuhan-Pole 2018). For instance, only 37 percent of agricultural households using fertilizers, while only 8 percent use improved seeds, and barely 1 percent have access to irrigation. Additionally, limited access to knowledge on modern farming practices, pest control, and climate-resilient techniques hinders technical efficiency and technological adoption. This keeps the sector largely focused on subsistence farming, with only 20 percent of production being commercialized and urban areas relying on imports for food consumption. The predominance of rainfed farming practices also mean that more erratic rainfall patterns and rising temperatures can severely impact the welfare of rural communities. These trends are already affecting production capabilities and farmers' welfare, most notably in the Northern Savanes region where heat stress and droughts tend to be most severe.

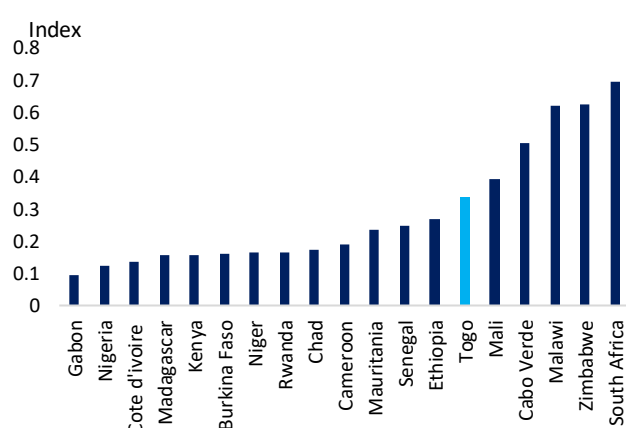
Figure 11. Labor productivity gains in agriculture are small compared with peers

Change in labor productivity in agriculture between 2010-2016 and 2017-2021



Source: World Bank

Figure 12. Togo's budget commitment in the agriculture sector is above the regional average but well below the size of the sector in the economy
Agriculture Orientation Index of public spending



Source: World Bank

Note: The AOI is defined as agriculture's share of public spending relative to its share in the economy.

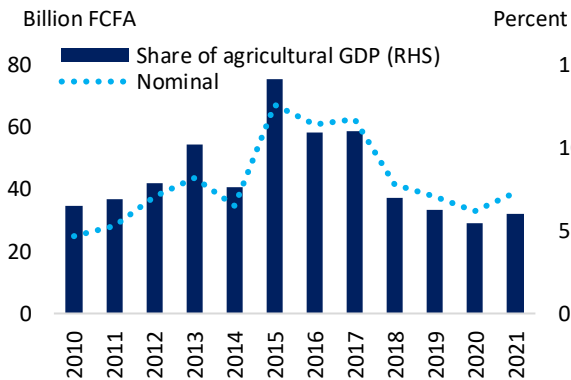
A crucial element in enhancing agricultural productivity is improving the provision of productive investments through more and better public spending in agriculture (Ali and Bayale 2024). This opportunity has been recognized by countries across Africa with the Comprehensive Africa Agriculture Development Programme (CAADP), which included a commitment to invest 10 percent of national public spending in agriculture (2003 Maputo Declaration and 2014 Malabo Declaration). While almost all countries in Africa are spending below the 10 percent target, country conditions and thus spending contexts differ widely. For instance, the spending target is arguably less meaningful for such countries as South Africa and Botswana, with small agricultural GDP shares in the overall economy. An alternative indicator of public sector budgetary commitment to agriculture is the Agriculture Orientation Index (AOI), defined as agriculture’s share of public spending relative to its share in the economy. An AOI value of 1 would indicate that the government spends a share of its budget on agriculture exactly proportionate to agriculture’s contribution to gross domestic product. This shows that Togo’s budget commitment in the sector sits above the average of countries in the region, but still well below the share of agriculture in GDP (Figure 14).

2. Level and efficiency of public investment in agriculture

Resources allocation to agriculture has been too low to meet government’s ambitions. Togo’s agricultural policy, PNIASAN (Programme National d’Investissement Agricole et de Sécurité Alimentaire et Nutritionnelle), sets out to double agriculture productivity between 2017 to 2026 by increasing farm mechanization rate from 6 to 25 percent; enhancing smallholder farmers' access to irrigation from 0.5 to 1.6 percent; raising the adoption rate of high-quality genetic materials (e.g. seeds, fry, chicks, and broodstock) from 4 to 35 percent; increasing access to fertilizers and pesticides from 4 to 10 percent; and doubling fertilizer consumption to 14 kg/ha. Despite these ambitions, public expenditure allocated to the agricultural sector has declined to 7.7 percent of agricultural GDP over the period 2016-21, down from an average of 8.8 percent during the 2010-15 programming period (Figure 15). As a percentage of overall public spending, the decline was less pronounced, dropping from 6.8 percent in 2010-2015 to 6.0 percent during the 2016-2021 period, but remained well short of the 10 percent target as part of the Maputo / Malabo commitments and at about half the share of agriculture value added in the economy, illustrating an important funding gap. This underperformance is partly explained for the later period by the completion of major projects, disruptions associated with the COVID pandemic, and increased spending in other sectors to address the economic and health crisis. The reliance on external financing also increased between the two programming periods (Figure 16), suggests a need for sustainable funding strategies to support the agricultural sector's growth and resilience.

Figure 13. Overall public spending in the agricultural sector has recently declined

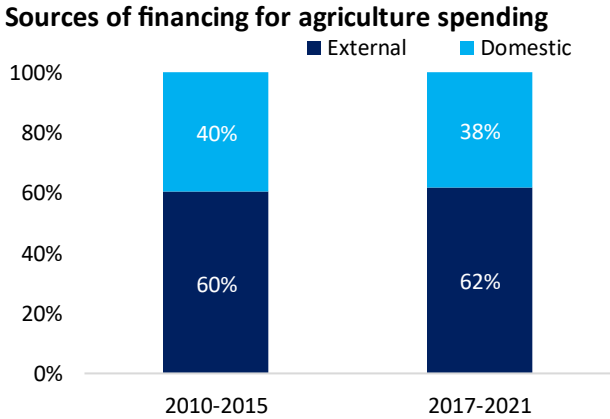
Public spending in agriculture



Source: World Bank

Figure 14. Sources of financing for public spending in agriculture are largely external

Sources of financing for agriculture spending



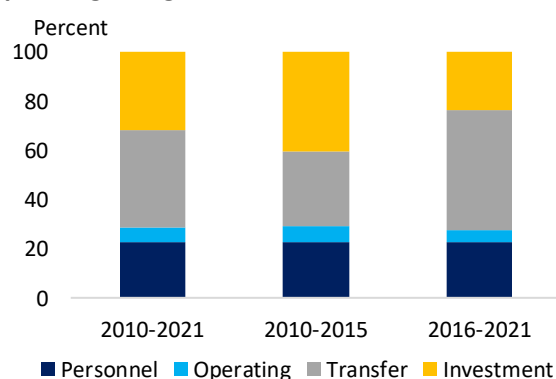
Source: World Bank

Mixed progress with public investment in agriculture reflects both low budget allocations and implementation challenges. Indeed, around three-quarters of domestic resources dedicated to agriculture were used during the 2016-21⁷ period to finance current expenditure, and most notably salaries and transfers, while less than one-quarter was dedicated to investment and only 5 percent to operating expenses (Figure 17). This left little margin to support investment projects and finance the provision of public services, particularly at the decentralized level. This was partly offset by the large share of capital spending financed by external resources and the fact that these generally include operating expenses for project implementation. However, externally financed projects are time-bound, which can create discontinuity in the provision of public services. Additionally, it removes responsibility from the State over time, and it may prove difficult to reintroduce budget lines for adequate funding for operating costs from internal resources.

The execution of investment projects in agriculture has been slow since the start of the new programming cycle in 2016. Regarding spending financed by internal resources, the execution rate for personnel expenses, operating expenses, and transfers and subsidies all exceeded 80 percent over the period 2016-21, but the execution rate dropped to 52 percent for capital expenses, down from 78 percent over the period 2010-2015 (Figure 18). This decline can be attributed to various factors such as poor performance by some service providers, particularly in the field of hydro-agricultural development, inefficiencies in the financial execution chain, and the impact of the COVID-19 pandemic in 2020. The execution of externally financed investments also faces specific constraints, including delays in receiving no-objection notices from certain partners, as well as late payments and difficulties in mobilizing counterpart funds, which slows project execution. Weaknesses in public investment management highlighted in the previous section have also impacted the allocative efficiency of spending in agriculture, including limitations in feasibility studies, availability of resources for the operation and upkeep of public assets, a lack of coordination between commitment, procurement, and cash flow plans, and insufficient consideration of climate vulnerabilities.

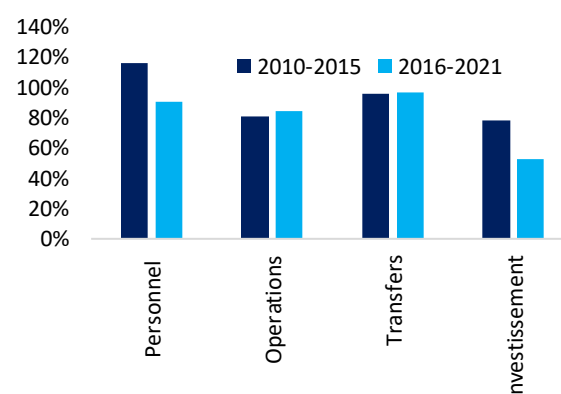
Figure 15. Domestically financed investment is low and declining

Breakdown of domestically financed public spending on agriculture



Source: BOOST

Figure 16. Budget execution rates are significantly lower for capital spending
Execution rate for public agriculture spending



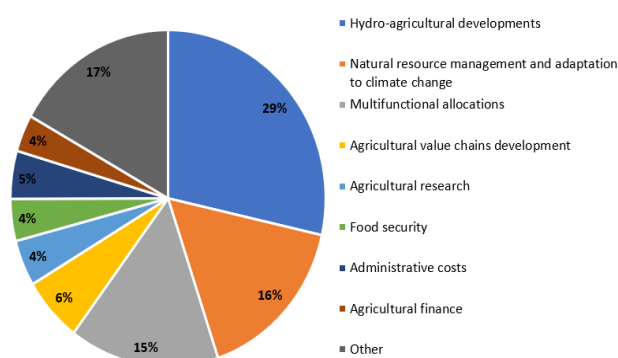
Source: BOOST

⁷ The latest BOOST data only covers up until 2021.

3. Progress towards government's objectives

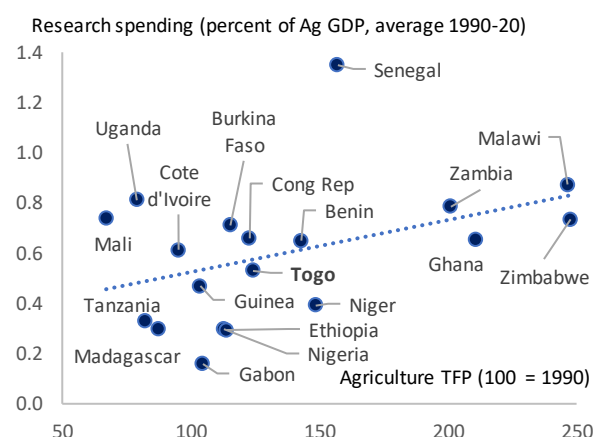
Public spending on agriculture has a strong focus on hydro-agricultural projects and natural resource management but shows very limit commitments on research and extension services, despite high returns. Over the 2017-2021 period, hydro-agricultural development accounted for around 30 percent of total spending in the sector, far exceeding other spending categories (Figure 19). Natural resource management and climate adaptation were allocated 16 percent of the budget, while support for agricultural value chains 6 percent, agricultural research, food security and finance 4 percent each, and extension services 3 percent. Other agricultural functions and multifunctional allocations⁸ (not disaggregated) accounted for 17 percent and 15 percent respectively. This spending structure highlights a lack of prioritization of research and extension services, which are generally found to have significant impacts on agriculture productivity⁹ (Figure 20). In fact, return on investments in basic and applied research across Sub-Saharan Africa are estimated to average about 36 percent, and 44 percent when research is combined with effective extension services, but with wide variations across existing studies (Pardey et al. 2016). This compares with estimated rate of returns on irrigation averaging about 7 percent for large-scale projects and 28 percent for small-scale projects in Sub-Saharan Africa (You et al. 2009), while input support measures like fertilizer subsidies appear to have average return on investment closer to 1.5 (Chirwa and Dorward 2013).

Figure 17. Agriculture research and extension services are underfunded in Togo
Functional breakdown of agriculture budget allocations over the 2017-2021 period



Source: DGB/MEF data

Figure 18. ...yet they are key to boosting productivity
Spending on research and productivity in agriculture



Source: World Bank

Weakness in the level, composition and efficiency of public spending are reflected in limited progress towards the government's objectives in the sector. Since the initiation of the new national agriculture and food security investment plan (Programme National d'Investissement Agricole et de Sécurité Alimentaire et Nutritionnelle, PNIASAN 2017-2026), agricultural yields have stagnated or declined, except for a marginal increase in maize yields. Efforts to improve seed access and fertilizer use have fallen short of targets, and fertilizer usage remains below recommended levels. The expansion of irrigated areas is significantly below planned goals, contributing only a small fraction to the total cultivated area. And only a small portion of the planned agropole areas has been developed to date. The completion rate of hydro-

⁸ They are made up essentially of allocations for projects to support youth employability in agriculture and projects to support grassroots development.

agricultural development work remains low due to insufficient public resources and the limited technical and financial capacity of companies involved. However, the expansion of Agricultural development zones (ZAAP) is promising, with farms in ZAAPs being significantly more efficient and productive than in non-ZAAP areas (Table 2). Looking ahead, improved water management, access to inputs, farm equipment, storage, preservation infrastructures, and better integration into value chains should further augment yield levels in ZAAPs.

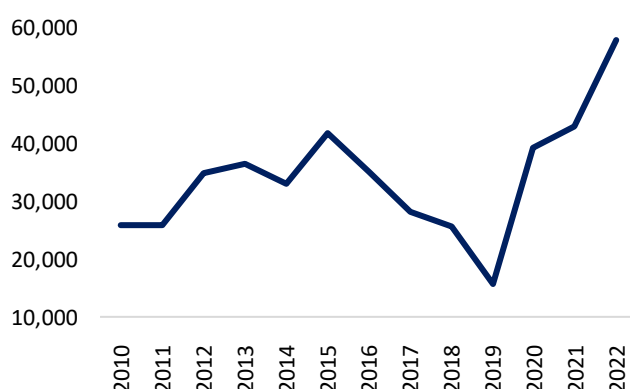
Table 2. Yields (in T/ha) for main crops in ZAAP versus non-ZAAP areas

Product	ZAAP	Non-ZAAP
Maize	2.1	1.3
Sorghum	1.1	0.9
Paddy rice	2.6	1.7
Cassava	9.6	4.0
Yam	25.0	9.0
Bean	0.7	0.5
Soybeans	1.3	1.02

Source: DSID, 2023

Despite uncertain benefits, the government's policy of facilitating access to subsidized fertilizers has absorbed an increasing share of public resources in recent years. State financial support for fertilizer subsidies is provided through the Centrale d'Achats et de Gestion d'Intrants Agricoles (CAGIA), a public body with legal personality and financial autonomy, with subsidy programs mainly focusing on food crops, which account for 70 percent of the fertilizer market, with cotton and other export crops accounting for the remaining 30 percent. Subsidized fertilizer quantities averaged 33,000 tons per year during the first phase of PNIASAN and increased to 35,000 tons per year over the 2017-2022 period, then jumping to 58,000 tons in 2022 amid efforts to comb rising fertilizer prices and reduced availability on international markets following the COVID 19 pandemic and the war in Ukraine (Figure 21). This came at increasing cost for public finances and other agriculture programs. Based on the quantities sold by CAGIA and the subsidized prices¹⁰, the subsidy is estimated to have averaged 3.5 billion FCFA over the period 2010-2021, increasing to 15 billion FCFA in 2022.

Figure 19. Volume of subsidized fertilizers (in tons) have increased amid rising prices



Source : <https://siar.uemoa.int/>

¹⁰ Actual expenditure on subsidies is difficult to trace in the budget data. In fact, they averaged 2.3 billion FCFA per year over the 2010-2015 period according to the budget data, compared with an estimate of over 5 billion based on the statistics provided. For the period 2016-2021, they amounted to 1.3 billion FCFA, compared with an estimate of around 2 billion per year.

A fertilizer market liberalization reform has been underway but with mixed successes. This liberalization reform allowed in principle the private sector to handle fertilizer orders and distribution, with subsidies granted to pre-targeted beneficiaries via an electronic platform. However, private operators are unable to supply the market due to their limited financial capacity and banks' reluctance to support them. The conflict in Ukraine has exacerbated market shocks, impeding the emergence of private operators. Additionally, CAGIA's withdrawal from the fertilizer order and the transfer of its warehouses to private players has been slow. It is up to the public authorities to speed up the liberalization of the fertilizer market by facilitating access to financing for private operators through the establishment of guarantee funds or lines of credit with financial structures. Additionally, fully implementing an electronic distribution system and phasing out price subsidies will reduce the inefficiencies of the current system. These important reforms must be accompanied by a strengthening of fertilizer quality control to guarantee their effectiveness.

The government supports the development of improved seeds through the organization of production and certification which also face significant challenges. Unlike fertilizers, there is no subsidy for improved seeds. Key institutions like the National Seed and Plant Council (CNSP) and the technical committee for the homologation of species and varieties suffer from an inadequate budget allocation and capacity constraints. In addition, seed producers' limited financial capacity means that they are unable to meet the costs of seed certification. Hence, the ongoing CAGIA's withdrawal from fertilizer distribution could free up resources to finance fertilizer quality control and strengthen the seed industry by supporting research and multiplication of resilient seeds, financing frameworks for concertation and homologation of species and varieties, strengthening analysis laboratories and human resources, and reducing the cost paid by seed producers for certification. Access to finance for smallholder farmers could also be supported by developing credit lines and guarantee funds specifically geared to respond to agriculture production needs.

Despite an increasing share, the budget for climate adaptation and mitigation in agriculture remains low, hindering efforts to strengthen resilience. Spending supporting natural resource management and adaptation currently cover 4 percent of financing needs, which are estimated at US\$350 million per year over the period 2020-30¹¹. Togo, like most other countries in the region, struggles to effectively tap into available climate funds due to limited capacity including for policy analysis and project formulation. This underscores the pressing need to build the capacity of the various actors to access and manage climate finance in Togo to boost the country's ability to receive and manage large-scale projects.

4. Key policy options

To boost agriculture productivity and resilience to shocks, key recommendations are identified along the four policy axes of the PNIASAN program:

- **Regarding Axis 1 on the Organization of Agricultural Space and Sectors**, hydro-agricultural development in ZAAP should be priorities, supported by a more careful selection of contractors and implementation of sanctions in case of technical failures. Access to improved agricultural inputs and equipment should also be facilitated in ZAAPs, and private leasing for warehouse management and contracting of hydro-agricultural facility maintenance to private service providers would be advised.
- **Regarding Axis 2 on Improving Productivity in the Agricultural Sector**, key recommendations include revitalizing agricultural research and advisory services through a clear national strategy, increased budget, better remuneration for researchers, improved financing and certification frameworks for the seed industry, reforms to fertilizer subsidies and strengthened quality controls. Establishing credit

¹¹ Green Climate Fund-Togo.

lines and guarantee funds with financial institutions should also help small farmers gain access to finance.

- **Regarding axis 3 on Improving Population's Resilience**, the main recommendation would be to strengthen the capacity of stakeholders to analyze fund policies and formulate projects, thereby facilitating access to climate funds.
- **Regarding axis 4 on Improving Governance in the Agricultural and Rural Sector**, key recommendations include enhancing budget execution by building the capacity of players in the financial chain, reducing processing times by donors to strengthen the execution of externally financed projects and promoting integrated projects to ensure more effective and coordinated interventions.

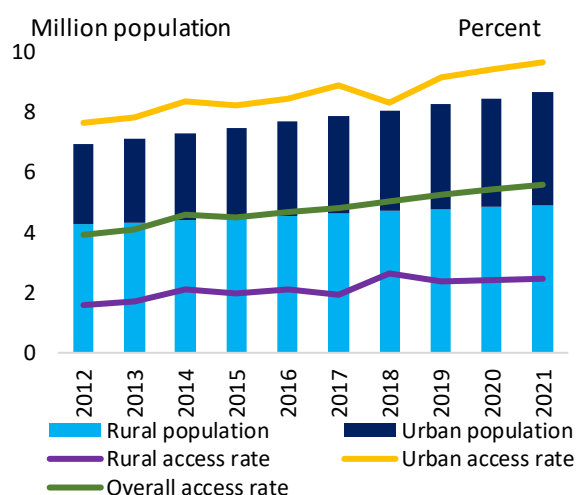
BROADENING ACCESS TO AFFORDABLE AND CLEAN ENERGY

1. Introduction

Access to electricity and the development of renewable energies are fundamental pillars of Togo's vision to modernize the rural economy and accelerate structural transformation. In a country where a large part of the population resides in rural areas and where agriculture plays a dominant role, addressing current low levels of rural electrification will be critical to boost the growth potential of the rural economy, and accelerate socioeconomic transformation more generally (Calderon 2009). In particular, supporting electricity-driven technologies in farming is key to ensuring a more productive and resilient agriculture sector (Amuakwa-Mensah and Surry 2022). Indeed, access to electricity empowers farmers to mechanize production, use electric processing equipment that add value to crops, support irrigation systems that mitigate dependence on unpredictable rainfall, allow for year-round cultivation and increased yields. This can significantly enhance income levels for farmers, contribute to the commercialization of food production and help meet growing food demand from urban areas. In addition to productivity gains, access to electricity can enable farmers to diversify their income sources by engaging in non-agricultural activities, such as setting up small businesses or agro-processing units. This, in turn, can help mitigate risks associated with climate variability, market fluctuations, and crop failures, making the agriculture sector more resilient to shocks. Rural electrification is also critical to reduce the digital divide, foster community empowerment, and strengthen climate adaptation as renewable energy sources like solar power reduce the reliance on fossil fuels and alternative lighting means such as kerosene lamps, which harm the environment and contribute to respiratory diseases.

Figure 20. Access to electricity has improved, but rural areas still have very low access rates

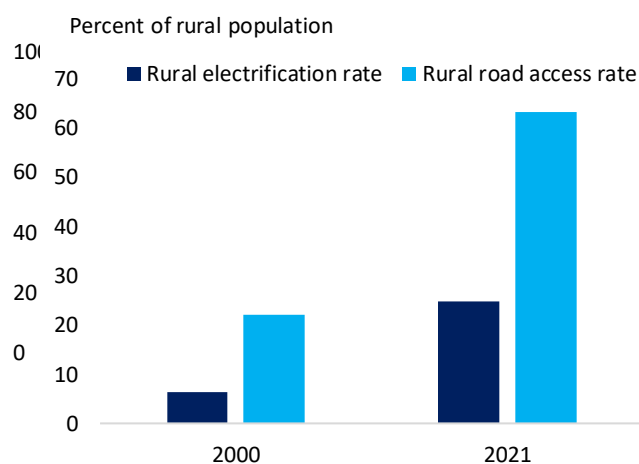
Access to electricity in Togo



Source: World Bank

Figure 21. Rural electrification in Togo made much slower progress than rural road access

Rural electrification and rural road access rates



Source: World Bank

Access to electricity in Togo has improved, but major disparities exist between urban and rural areas. The national electrification rate has increased steadily from 39 percent in 2012 to 56 percent in 2021 (Figure 22). However, this uptick was mainly driven by electrification efforts in urban areas, where access rate reached 96 percent in 2021, from 76 percent in 2012. Conversely, despite government's efforts, rural areas continue to face limited access to electricity, with approximately 3.7 million individuals without access to electricity in 2021. Unlike urban areas, progress in rural electrification has been sluggish, only reaching 25 percent in 2021, up from 16 percent in 2012, pointing to one of the largest rural-urban divides among WAEMU countries. This places Togo in the bottom decile of rural electrification rates globally

(Figure 23). Moreover, even as 60 percent of the rural population lives within 2 km of a road, only 24 percent of these roads are electrified, as installation requires substantial investments that are often unattainable for resource-limited rural areas. To achieve the government's target of reaching universal access by 2030, about 100,000 households would need to be connected per year, mostly in rural areas. This would require considerable investments that the National Utilities Company CEET and the Togolese Agency for Rural Electrification and Renewable Energies AT2ER are not in position to mobilize currently.

In the meantime, Togo continues to rely heavily on imports from neighboring countries for its energy supply. Strides in off-grid renewable energy production have contributed to lowering energy imports to 55 percent of overall consumption in 2021, down from 62 percent in 2017. Despite progress, the continued high dependence on imports from Nigeria and Ghana via the Communaute Electrique du Benin (CEB) interconnectors makes Togo vulnerable to price fluctuations, regional energy sector crises, posing challenges in terms of energy security and sustainability. Additionally, elevated debt to utilities in exporting countries hinders the reliability of these imports.

Togo is actively exploring alternative energy sources, investing in hydroelectric and solar projects to reduce its dependence on imports. The development plan for renewable energies hinges on the attractiveness of private investment and the evolution of public-private partnerships, but these have been undermined by the lack of viability of the public utility company. For private investors and stakeholders in public-private partnerships, CEET's financial health, regulatory stability, and transparency are crucial. If these elements are not robustly established, it could result in reduced investor confidence, leading to cautious, limited, or withdrawn investments. This hesitancy can be particularly pronounced in an environment where operational costs, market structures, and revenue streams are not clearly balanced or present risks that are deemed unacceptable. Consequently, the ambition of Togo to transition towards a more sustainable and secure energy supply not only necessitates the initiation of projects but also a foundational reassessment and strengthening of the economic viability of the entire power sector to foster and sustain the investor confidence essential for these projects' success.

2. Level and quality of public investment and financial viability of the power sector

Current budget allocation for Togo's electricity sector is inadequate, necessitating increased financing and tariff adjustments to meet investment needs. Despite the increase in the budget allocation for public investment in the electricity sector to about 0.4 percent of GDP in 2021 (Table 3), this allocation remains insufficient to cover investment needs, which are estimated at about 0.7 percent of GDP annually, while spending efficiency is affected by weaknesses in public investment management, including limitations with strategic planning, integration of recurring costs for upkeep and maintenance, public procurement and PPP management. To meet significant investment needs, financing commitments would need to increase substantially, but the government's multiannual budget plan for 2023-2025 does not foresee such increase, underscoring the urgent need to mobilize additional resources. Between 2022 and 2025, the total allocated budget is \$381 million. However, to fully implement the master plan, it is necessary to mobilize \$669 million, representing a funding gap of 43 percent. The poor financial health of the national utility CEET is another major hindering factor for investments in new infrastructure, the upkeep and maintenance of existing infrastructure, and improved service quality. Underpricing of services remains the predominant source of losses for the CEET, with the average selling price currently standing at about \$0.19/kWh, while the estimated cost of service is around \$0.24/kWh. Considering the improvement in the energy mix, baseline performance, and investment needs, tariffs would likely need to increase between 16 to 18 percent from current levels to restore CEET's financial equilibrium (Figure 24).

The differentiated volume-based tariff system in Togo disproportionately benefits wealthier households and concentrates in urban areas, to the disadvantage of rural areas. The government employs Volume-Differentiated tariffs (VDT) systems within the electricity and water sectors, offering reduced tariffs to low utility-consumption households. A VDT system aims at providing the most substantial subsidies to those

households facing greater economic challenges. A prepaid service with a uniform tariff is also used, providing the same subsidy to all households connected to the grid, regardless of their consumption levels or income. Across all deciles, electricity subsidies represent between 0.1 to 0.3 percent of household budgets. Electricity subsidies are significantly skewed towards the wealthiest with the top 10 percent receiving 39.5 percent of total spending on these subsidies, while the bottom 10 percent receive merely 2 percent of total spending (Figure 25). The Grand Lomé region is the recipient of 61.5 percent of these subsidies, while other urban and rural areas get 17 percent and 21 percent, respectively, revealing pronounced geographic disparities.

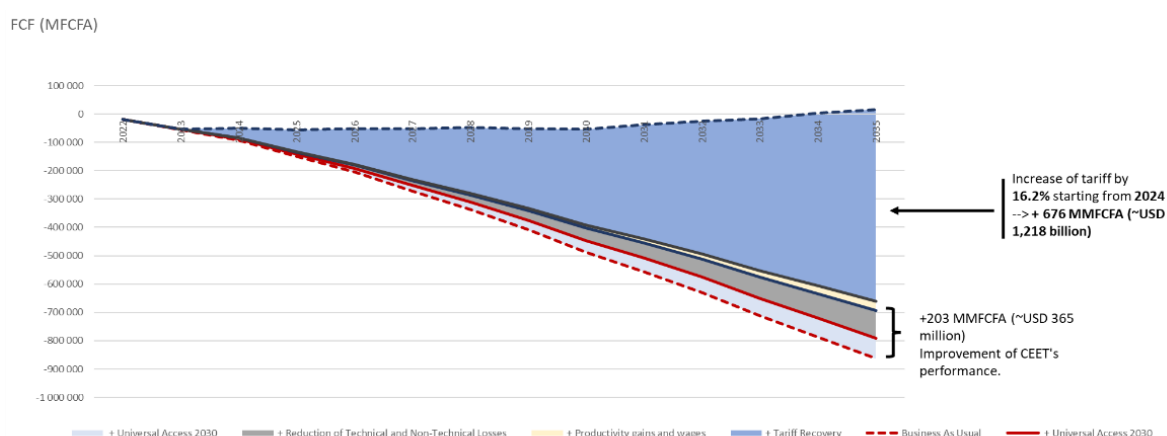
Table 3. Public investment in the electricity sector

(In billion FCFA)	2017-2020			2021		
	Budget	Execution	Execution rate	Budget	Execution	Execution rate
Public Investment	70.0	63.4	90%	33.4	23.6	71%
Internal Funding	4.5	3.1	68%	1.4	1.5	110%
External Funding	65.5	60.3	92%	32.0	22.1	69%
%GDP	0.2	0.2		0.4	0.3	

Source: BOOST and CEET capital expenditure

Note: The lack of data on investments financed with external resources at the BOOST level (amounts, sectoral allocations, executions) makes the analysis less complete. To derive reliable estimates of the budget execution rate for externally financed investments, major projects such as the construction of the 330 KV Ghana-Togo-Benin interconnection line, the rehabilitation of the Nangbeto hydroelectric plant, and the Blitta solar power plant construction were excluded, as well as CEET's investments funded by its own resources.

Figure 22. Estimated contributions to restoration of CEET's financial equilibrium

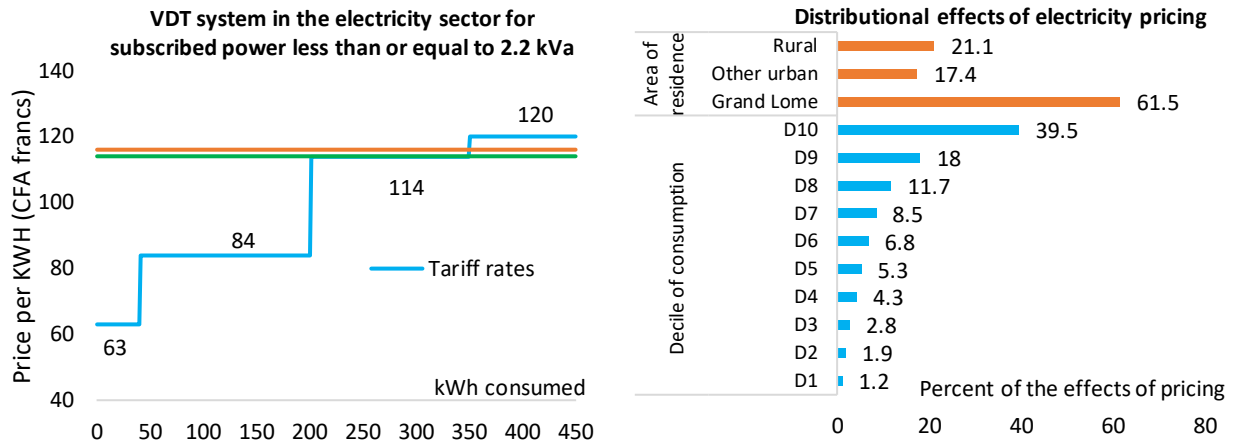


Source: World Bank

Despite progressive pricing, higher income individuals predominantly benefit from subsidized tariffs.

This is due to the broad definition of tariff bands, which allow even high consumption levels to receive subsidized tariffs. In particular, the category lower tariff “Conventional Domestic” category accounts for 82 percent of all domestic clients, of which only 24 percent consume less than 40 kWh. Additionally, a prepaid service with a uniform rate is also employed, offering the same subsidy to all households connected to the network, regardless of their consumption level or income. To reduce this disparity, it has been recommended to (i) lower the maximum eligibility threshold for the social tier (the tariff study recommends a threshold of 20 kWh) and to review the application terms (selectivity) to reconsider its extension to all customers, (ii) segment consumption above 20 kWh into two bands ('0-120' and '120 kWh and above'), (iii) ensure better alignment of the tariff structure for prepaid clients with that of traditional metered clients, and (iv) revise rates upwards to fully cover service costs.

Figure 23. *The current tariff structure in Togo disproportionately benefits wealthier households*



Source: CEET, World Bank

Source: EHCVM 2021/22, Authorities, World Bank

Note: The chart shows Togo's Volume-Differentiated tariffs (VDT) systems. The government employs this tiered pricing for electricity and water to subsidize low-income users.

AT2ER, which oversees implementing Togo's rural electrification strategy, is also facing serious challenges. The primary mission of the AT2ER is to stimulate rural electrification while increasing the share of renewable energy in the installed production capacity. The AT2ER is also responsible for research and fundraising, developing funding mechanisms for rural electrification initiatives, and managing all procurement procedures for rural electrification and renewable energy projects. Nevertheless, the company faces a lack of resources, impeding its activities. In 2020, the operating budget represented only 1.5 percent of the total budget (509 million FCFA or about USD 0,9 million), far below the threshold of 15 percent established by law. While the investment budget reached 33 billion FCFA (about USD 59 million) that year, only 67 percent of it was actually disbursed. AT2ER's budget restrictions have a direct impact on its ability to carry out its mission.

The corporate governance of entities such as CEET and AT2ER has several weaknesses. There is insufficient oversight by the SOE ownership institution and a lack of financial transparency due to the absence of systematic publication of annual audited financial statements. Additionally, there is no clear accounting separation of commercial activities and public service obligations, complicating the assessment of the true costs of services. Performance agreements lack realism and clarity regarding objectives and responsibilities. Furthermore, the management and board of directors do not include enough independent directors with sector expertise, limiting informed decision-making. Finally, the absence of performance-related incentives, such as bonuses and sanctions, reduces motivation to achieve high performance and improve operations.

3. Key policy options

To accelerate rural electrification efforts and improve electricity services, the following of the following priority actions are identified:

- First, the institutional and regulatory framework could be improved to increase the financial autonomy of the AT2ER, strengthen the role of the sector regulator ARSE, and specify network access conditions for decentralized renewable energy production.
- Second, the performance and accountability mechanisms for the CEET and AT2ER should be reinforced, setting clear performance goals, building capacity, mainstreaming technological improvements, and increasing transparency.

- Third, an optimized tariff structure for electricity should be decided to better protect the poor, ensure sustainable revenues for the CEET, support decentralized renewable energy solutions and ensure industrial competitiveness.
- Fourth, the structure of the Tinga fund for rural electrification should be reviewed, considering the pilot phase results and define the modalities for integrating the existing CIZO subsidy and its allocation to the most vulnerable households.
- Finally, the security of gas supply for thermal power plants should be reinforced by negotiating more favorable tariffs and considering the possibility of establishing hedging contracts for gas price and exchange rate fluctuations.

STRENGTHENING RURAL ROAD CONNECTIVITY

1. Introduction

A well-developed and maintained rural road network is crucial to unleash economic opportunities in rural areas. Improved road connectivity directly enhances access to markets, allowing farmers to efficiently transport their products, hence reducing post-harvest losses, and ensuring better market prices (Dorosh et al. 2012; Tunde and Adeniyi 2012; Calderón, Cantu, and Chuhan-Pole 2018; Iimi 2021; Wiegand et al. 2023). Indeed, all-year access to an efficient and well-maintained rural road network allows for more predictable, faster and cheaper transport of crop production to markets and processing facilities, improve access to inputs and extension services, which empower farmers to adopt best practices, improve yields, and participate in more profitable agricultural activities. Furthermore, enhanced rural road connectivity fosters market access and diversification. Previously isolated rural communities gain access to a wider range of markets, allowing farmers to sell their produce beyond local markets and potentially connect with higher-value markets in urban centers. This diversification reduces dependence on single crops and fosters agricultural production that caters to a wider range of consumers. This stimulates agricultural diversification and encourages the adoption of more modern and capital-intensive farming practices, which are crucial to boost productivity and resilience to climate shocks. Additionally, reliable roads can attract investments into local industries, creating employment opportunities beyond agriculture. These developments can lead to a more dynamic rural economy, reducing poverty and fragility risks. Furthermore, better road infrastructure improves access to education and healthcare, which could contribute to closing the gap in human capital development between rural and urban areas. Overall, a robust rural road network is a linchpin in Togo's efforts to shift from a predominantly agrarian rural economy to a more diversified and resilient one.

Enhancing and maintaining Togo's road sector requires substantial investment and strategic resource mobilization (including private and external sources). Togo has a total road network of approximately 12,419 km, including 3,152 km of national roads - forming the backbone of the network - and 7,384 km of secondary roads essentially made up of unclassified rural tracks (Figure 26). While the average transport speed between cities is close to the average of low-income countries, the rural road access rate tends to be somewhat higher, which partly stems from the fact that Togo is a relatively small and narrow territory which allows small number of feeder roads to connect significant parts of rural areas (Figure 27). In fact, 63 percent of the rural population in Togo lives within two kilometers (equivalent to a 20-25 minute walk) of an all-weather road, which is similar to neighboring Ghana (64 percent) and slightly better than Benin (59 percent), but still lower than aspirational peers like Bangladesh (67 percent), Morocco (75 percent), Rwanda (80 percent), or Vietnam (85 percent). This said, many rural roads are unpaved and poorly maintained, rendering them vulnerable to natural hazards. During the rainy seasons, rural roads and tracks often become impassable, isolating communities and hindering the movement of people and goods. Increasing the rural road access rate from current levels would require mobilizing significant financial resources. In fact, based on existing rural roads condition, needed improvements, and estimated road construction costs, increasing the access rate to 75 percent of the rural population could cost more than 1 billion USD, and require annual maintenance costs amounting to 2.2 percent of GDP in (Figure 28). If financed solely by own resources, this could put tremendous pressure on the government budget at time when it must prioritize consolidation efforts. This highlights the importance of project prioritization, mobilization of external concessional resources, and private sector participation.

Building climate-resilient roads in Togo is crucial to mitigate economic losses from climate-related damages. Togo frequently faces extreme weather events like heavy rains, flooding, and high temperatures, that cause roads and bridges to deteriorate, which in turn drives up maintenance costs and

disrupting transport. The cumulative costs due to climate related road and bridge damages could reach between \$249 and \$536 million over the period 2041-2050. This underscores the urgent need for climate-resilient roads and bridges to ensure continued transport network functionality and minimize economic disruptions. However, substantial upfront costs associated with such infrastructure, coupled with existing budgetary constraints, make such investment difficult. Additionally, technical expertise and access to advanced materials for building climate-resilient roads are limited. Furthermore, a long-term strategy is vital, encompassing not only construction but also sustainable maintenance practices while integrating environmental considerations. Striking the right balance between the economic benefits of resilient infrastructure and potential ecological impacts, such as deforestation, is crucial. If newly paved rural roads lead to deforestation in an area extending 25 meters on each side of the road (minimum impact, mainly due to road works), about 2 percent of the forest cover could be impacted in Togo (Figure 29). This emphasizes the importance of addressing financial, technical, environmental, and policy-related challenges and prioritizing rural road investments according to their economic and social benefits as well as their potential environmental impacts.

Figure 24. *The road network in Togo is substantial, ...*

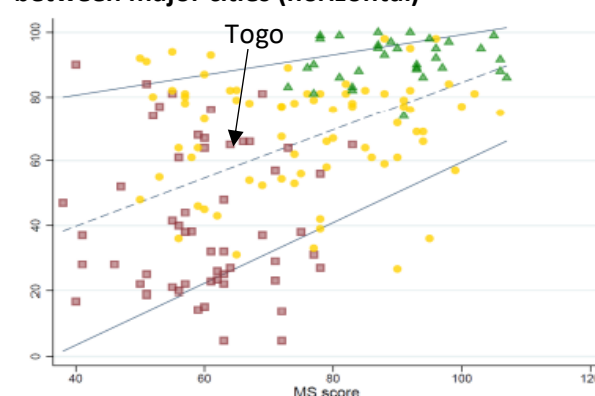
Road network in Togo



Source : Worldometers

Figure 25. *... but travel speed between cities remains low compared to peers*

Rural road access rate (vertical) and travel speed between major cities (horizontal)

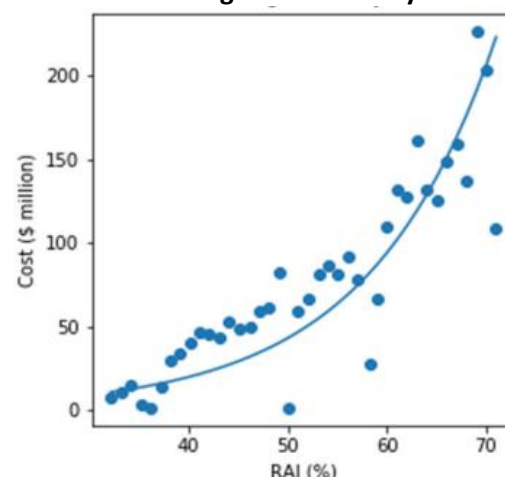


Source: Moszoro and Soto (2022)

Note: Round red squares represent low-income countries, yellow circles represent emerging market economies, and green triangles represent advanced economies

Figure 26. *Improving rural access rate in Togo will become increasingly costly*

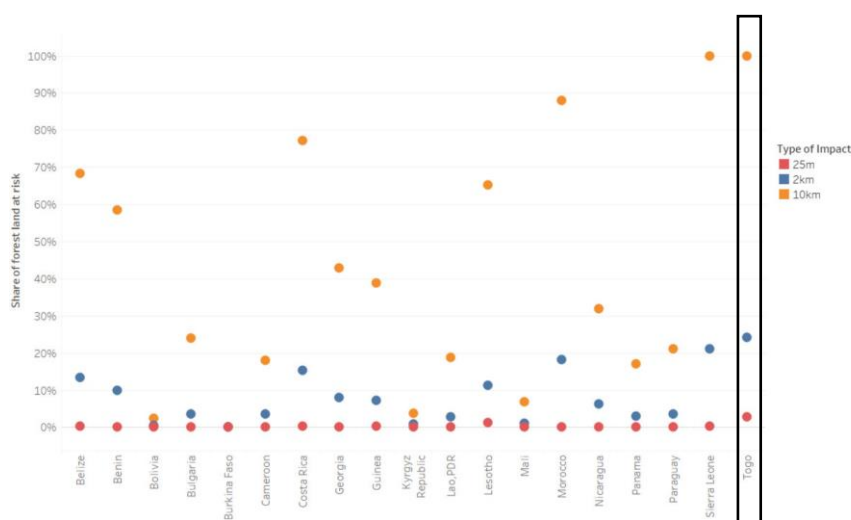
Cost of increasing rural access by 1% in Togo



Source : Mikou et al. (2019)

Figure 27. Road paving poses a significant threat to forest cover

Impact of rural road paving on the forest cover



Source: Mikou et al. (2019)

2. Level and quality of public investment in the road sector

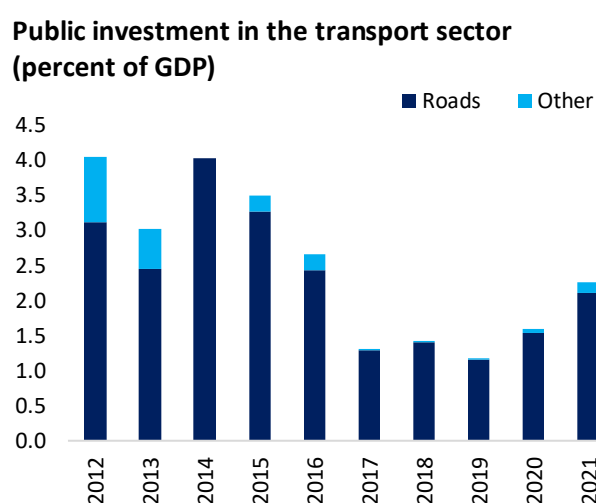
The government has recognized the importance of improving road infrastructure to enhance trade competitiveness and support rural development, yet rural-urban gap remains high. The past decade has seen substantial public spending to support the government's road infrastructure program, with public investment in the sector averaging 2.5 percent of GDP per year. Capital spending in the road sector was particularly strong over the period 2012-16, reaching a peak of 4 percent of GDP in 2014, as the government aimed to boost connectivity and make Togo a regional trade hub (Figure 30). Nonetheless, it dropped to an annual average of 1.3 percent of GDP between 2017 and 2019, in part reflecting fiscal consolidation efforts amid debt sustainability concerns, before rebounding to 2.1 percent in 2021.¹² The government has prioritized the rehabilitation and maintenance of existing road infrastructure, with a particular focus on restoring the viability of the transit corridors, rehabilitating the main trunk roads and priority feeder roads to improve connectivity and accessibility (Figure 31). This supported the steady increase in the proportion of national roads in good condition to 46.4 percent in 2021, and an increase in the length of national roads being maintained annually, from 1337.6 km in 2020 to 1698 km in 2021, and an estimated 1685 km in 2022. A specific budget dedicated to rural roads only started in 2018 and remained very low compared to other investments in the sector. The creation in October 2020 of a Ministry in charge of Rural Roads ("Désenclavement et pistes Rurales") with an autonomous budget to ensure the management and development of the network should help improve the management of limited budget resources.

Togo relies mostly on domestic public resources for road sector funding, with external funding focused on rehabilitation and maintenance. From 2012 to 2021, external resources financed about 36 percent of capital expenditures in the road sub-sector. However, this share significantly increased during the fiscal consolidation episode of 2017-2019, reaching 64 percent in 2018, partially offsetting the reduction in domestic spending. External financing mainly funded rehabilitation and maintenance road projects and was primarily supported by external development partners. Private sector participation remains sluggish, constrained by weaknesses in the regulatory framework, high costs, and limited capacity and expertise.

¹² Public spending in the Transport sector/Road Transport sub-sector are identified using the tagging method. Public spending in transport includes public spending related to roads, aviation, aeronef, airport, and maritime business.

Over the past decades, the only PPP registered in the sector was directed to the maritime transport sub-sector. A relatively high execution rate of domestically financed investment projects in the road sub-sector denotes the government's strong engagement in the sector, but road infrastructure gaps remain substantial and maintenance efforts are insufficient. Weaknesses in the public investment management highlighted earlier apply to the road sector as well, notably in terms of the coherence of strategic objectives, budget projections and actual commitments, insufficient account of recurring costs and resources needed for the operation and upkeep of assets over time, barrier to efficient procurement and PPP participation, and a lack of assessment of climate vulnerabilities in the project appraisal and selection process.

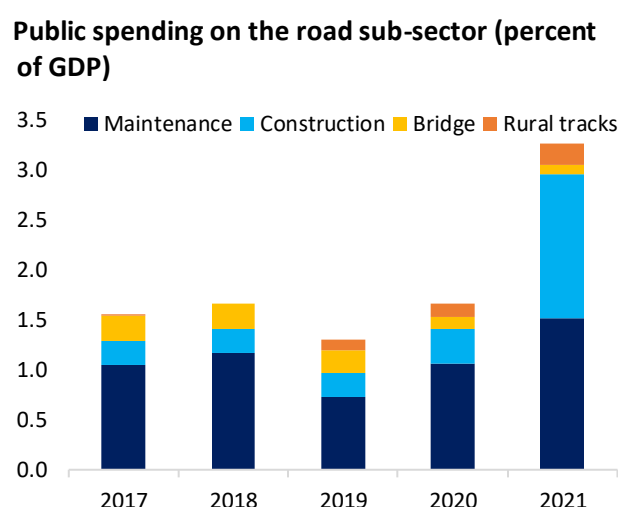
Figure 28. *Public investment in the transport sector increased since 2020, but from low levels*



Source: BOOST, MFMOD

Note: Tagging method was used to identify spending in Transport sector. Transport sector includes covers spending related to aviation, aeronef, airport, maritime business.

Figure 29. *Road construction and maintenance efforts were the main driver*



Source: BOOST, MFMOD

3. Capacity to support road maintenance and rehabilitation

Estimated costs of maintenance and rehabilitation highlight a significant funding gap, underscoring the need for complementary resources, including fuel subsidy reforms. Estimations of the number of kilometers of deteriorated roads by category and regions in 2021 are used to calculate the associated costs of maintenance and rehabilitation.¹³ Based on these estimates, maintenance needs would amount to 0.9 percent of GDP and rehabilitation needs would reach 5.6 percent of GDP. Additionally, it is anticipated that the impact of climate change will contribute to an escalation in both maintenance and rehabilitation costs, which will further increase the financing gap in the sector. SAFER's overall resources reached only 1 percent of GDP in 2022, with 0.25 percent of GDP originating from its own resources (tolls and weighing station), 0.33 percent from transfers by the State (excise duties and TVM) and the rest being unspent revenues carried over from previous years. With these resources, the SAFER spent 0.86 percent of GDP in maintaining about 2,707 km of the national roads in 2022. SAFER's budget for 2023 includes priority road maintenance, multi-year road maintenance contracts, and the commissioning and operation of two new tolls (Kémériida and Atétou). The reinforcement of axle load control will also benefit from the commissioning of an additional two new tolls (Tsévié and Mango), as well as mobile axle scales. While

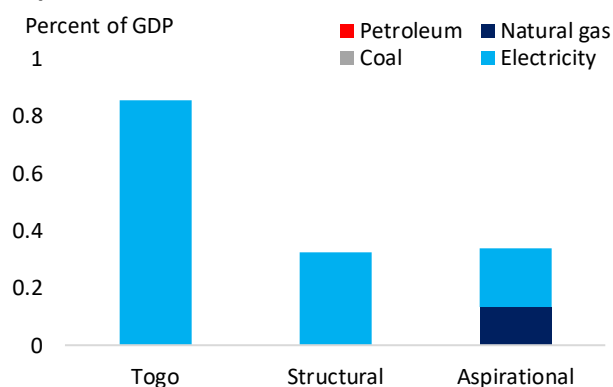
¹³ Estimated maintenance costs are obtained by applying the reference unit prices available from the DER to the quantities involved, and the rehabilitation requirement is obtained by applying five hundred million euros per kilometer to the gross rehabilitation length.

noteworthy, these resources remain insufficient to meet large investment needs and could be complemented by additional resources associated with fuel subsidy reforms, amongst others. SAFER also faces important governance challenges that hinder its effectiveness, including its issues of autonomy, transparency and accountability, procurement processes, and lack of community engagement.

Phasing out fuel subsidies could potentially generate significant revenue that could be redirected towards extending and maintaining the country's rural road network. Togo currently spends a substantial portion of its budget on fuel subsidies, which aim to keep fuel prices artificially low for consumers (Figure 32). For instance, in 2022, fuel subsidies cost the government around 1 percent of GDP, which is higher than the domestic budget for social protection programs. These subsidies are highly regressive since 40 percent of their direct benefits accrue to the richest 10 percent in Togo. There are often ineffective at stimulating activity and are harmful to the environment. If Togo considers phasing out fuel subsidies, it will need to consider compensatory measures to reduce its impact. One option is to provide targeted subsidies to low-income households through a cash transfer program or by providing discounted fuel cards to qualified households. The remaining cost savings could be allocated to supporting road construction and maintenance investments or support public transportation.

Figure 30. Large fossil fuel prices in Togo hamper the efficient allocation of economic resources

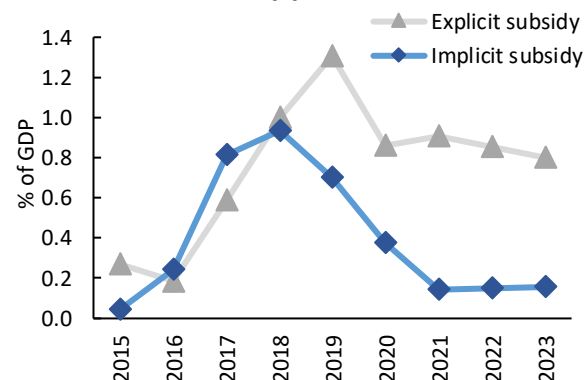
Explicit fuel subsidies, 2010-2022



Source: IMF, Parry, Black, and Vernon (2021)

Note: Explicit subsidies occur when the retail prices are below fuel's supply cost (domestic production cost).

Total fuel subsidies by year



Source: IMF, Parry, Black, and Vernon (2021)

Note: Implicit subsidies refer to undercharging for environmental costs and foregone consumption taxes.

4. Key policy options

Improving rural road connectivity will require a comprehensive approach encompassing improvements in technical and operational capacity, public procurement, resource mobilization, and private sector engagement.

- Priority interventions include upgrading the road assets management system with regular data collection for informed decision-making, refining the institutional framework for better coordination and investment prioritization, and conducting climate risk analysis to prioritize climate-resilient road projects.
- Updating rural planning strategies for resilience and sustainable development is also needed, together with additional resources for road maintenance and rehabilitation, including through levies on car insurance, taxes on spare parts, and fuel subsidy reforms, as well as enhanced technical and operational capacity and better governance and transparency at SAFER.

- Streamlining public procurement is also critical to reduce delays and improve the selection of competent companies, while leveraging private-public partnerships (PPPs) and external donor funds should help fill infrastructure gaps.

REFORM PRIORITIZATION

A small number of critical reforms could help accelerate progress towards rural development and structural transformation. While this report identifies a range of policy and institutional reforms that could help leverage limited public resources to untap the potential of the rural economy, a smaller set of priority interventions is identified based on five criteria: (i) the impact of the proposed measure on spending efficiency; (ii) the degree of urgency of the reform; (iii) the extent to which the intervention is necessary for other measures to yield benefits; (iv) the feasibility and appetite of stakeholders for the selected reforms and (v) their degree of complexity given existing capacity constraints. Using this framework, priority interventions are identified in Table 4 along four key areas.

Regarding public investment management, improvements in allocative efficiency, climate readiness and public procurement are particularly important. Key priorities include further regulatory and institutional improvements to clarify the role of various actors, alongside operational improvements for selecting and managing projects, and aligning them better with broader development and climate objectives. Greater emphasis should be put on timely and budget-conscious project implementation, integrating recurring maintenance costs, and developing robust tools to support climate risk assessments and capacity to plan and implement climate resilient infrastructures. Additionally, the private sector should be mobilized for infrastructure developments through more transparent and effective preparation of PPP projects, simplified and more transparent procurement, and fair competition through strengthened regulatory frameworks.

In the agriculture sector, efforts should concentrate on measures able to boost productivity and strengthen resilience to climate shocks. Key recommendations include more stringent contractor selection for hydro-agricultural development, revitalizing agricultural research and extension services, enhancing the seed industry's financing and certification framework, and reforms to fertilizer subsidies. Strengthening capacity and improving budget execution and project coordination are also key in an environment characterized by binding fiscal constraints.

To accelerate rural electrification, priority actions include optimizing the tariff system and improving governance at the CEET and AT2ER. These include enhancing the performance and accountability for the two institutions with clear goals, capacity building efforts, and more financial and operational transparency, clarifying network access for decentralized renewable energy and optimizing the tariff structure for electricity. Reviewing the Tinga fund's structure to boost rural connectivity and merging it with the CIZO subsidy for vulnerable households are also recommended.

To boost rural road connectivity, priority interventions include upgrading the road assets management system, refining the institutional framework, and helping to develop a more climate resilient road network. This involves developing the capacity to identify climate vulnerabilities, upgrading rural planning strategies, use more modern techniques for road building and maintenance, and mobilizing additional resources for road maintenance and rehabilitation, including through levies on car insurance, taxes on spare parts, and fuel subsidy reforms. Technical and operational capacity at the SAFER should also be strengthened together with clearer performance goals, more accountability and transparency. Streamlining public procurement is also critical to reduce delays and improve the selection of competent companies, while leveraging private-public partnerships (PPPs) could help fill financing and technical gaps.

Communication and public engagement are critical for the success of these reforms. Informing the population about the need and benefit of reforms is key to ensure the necessary support for change and increase the likelihood of policy success. Building coalitions of reform supporters, strengthening communication, and supporting third-party voice could all contribute to reinforcing policy buy in.

Community-driven solutions could also help deliver improvements in infrastructure quality at the local level, particularly in more remote areas where the state presence is more limited.

Table 4. Prioritization of policy interventions

No.	Policy Interventions	Time Horizon	Complexity
A. Public investment management			
1	Systematically cost sectoral strategies and ensure that they are accompanied by concrete targets to be achieved.	Short term	Moderate
2	Systematically include upkeep and maintenance costs in investment budgeting and systematize the production and publication of feasibility studies for investment projects.	Short term	Moderate
3	Develop and implement an integrated geographic information system for public investment planning.	Short term	High
4	Develop rules and procedures to assess climate risks and include them in the appraisal, selection, design and implementation of projects.	Medium term	High
5	Simplify public procurement processes to ensure more competitive bidding and increase transparency.	Short term	Moderate
B. Agriculture productivity and resilience to shocks			
6	Design and implement a targeted voucher system that enables smallholder farmers to access fertilizers and certified seeds, ensuring both affordability and timely access to quality agricultural inputs.	Short term	Moderate
7	Promote private leasing and PPPs for the development of low land, mechanization equipment and services, and irrigation schemes.	Medium term	Moderate
8	Develop agricultural research and extension strategies and increase budget allocations for both.	Medium term	Moderate
9	Improve private sector participation in fertilizer distribution and seed certification while reducing the role of public sector and strengthening fertilizer and seed quality control.	Medium term	Moderate
10	Improve access to finance by encouraging financial institutions to develop financial products aligned with the agriculture production cycle.	Short term	Moderate
C. Rural electrification and renewable energy			
11	Set clear performance goals for the CEET and AT2ER and regularly publish comprehensive reports on their operations, finances, and performance.	Short term	Low
12	Clarify network access for decentralized renewable energy to encourage their adoption by the private sector	Short term	Low
13	Optimize tariff structure for electricity to better protect the poor, ensure sustainable revenues for the CEET, and support decentralized renewable energy solutions.	Short term	High
14	Review the structure of the Tinga fund for rural electrification and define the modalities for integrating the CIZO subsidy for most vulnerable households in the fund.	Short term	Moderate
D. Rural road connectivity			
15	Enhance road assets management systems and ensure routine data collection on road inventory and traffic data for better prioritization of investment needs.	Short term	Moderate
16	Conduct systematic climate risk analysis and vulnerability for the existing transport projects and update rural planning strategies to prioritize resilient infrastructures.	Medium term	High
17	Strengthen public procurement processes to eliminate administrative delays and retain high-performing companies.	Short term	Moderate
18	Phase out fuel subsidies and use proceeds to finance the extension and maintenance of the rural road network.	Medium term	High
19	Increase the technical and operational capacity of SAFER and ensure regular disclosure of financial information and performance data.	Medium term	Moderate

REFERENCES

- Ali, Essossinam, and Nimonka Bayale. 2024. "Impact of Public Agricultural Investment on Crops Production, Households' Welfare, and Employment Generation Opportunities in Togo, West Africa." *The European Journal of Development Research* 36 (1): 161–193.
- Amuakwa-Mensah, Salome, and Yves Surry. 2022. "Association between Rural Electrification and Agricultural Output: Evidence from Sub-Saharan Africa." *World Development Perspectives* 25 (March): 100392.
- Ashkenazy, Amit, Tzruya Calvão Chebach, Karlheinz Knickel, Sarah Peter, Boaz Horowitz, and Rivka Offenbach. 2018. "Operationalising Resilience in Farms and Rural Regions – Findings from Fourteen Case Studies." *Journal of Rural Studies* 59 (April): 211–21.
- Azolibe, Chukwuebuka Bernard, and Jisike Jude Okonkwo. 2020. "Infrastructure Development and Industrial Sector Productivity in Sub-Saharan Africa." *Journal of Economics and Development* 22 (1): 91–109.
- Bahal, Girish, Mehdi Raissi, and Volodymyr Tulin. 2018. "Crowding-out or Crowding-in? Public and Private Investment in India." *World Development* 109 (September): 323–33.
- Calderon, C., and L. Serven. 2004. "The Effects Of Infrastructure Development On Growth And Income Distribution." *Annals of Economics and Finance* 15 (2): 521–34.
- Calderon, Cesar. 2009. "Infrastructure And Growth In Africa." *World Bank, Policy Research Working Papers*.
- Calderón, César, Catalina Cantu, and Punam Chuhan-Pole. 2018. "Infrastructure Development in Sub-Saharan Africa: A Scorecard." *World Bank Policy Research Working Paper* 8425.
- Calderón, César, and Luis Servén. 2004. "The Effects of Infrastructure Development on Growth and Income Distribution." SSRN Scholarly Paper. Rochester, NY.
- Calderón, César, and Luis Servén. 2010. "Infrastructure and Economic Development in Sub-Saharan Africa." *Journal of African Economies* 19: i13--i87.
- Cavallo, Eduardo, and Christian Daude. 2011. "Public Investment in Developing Countries: A Blessing or a Curse?" *Journal of Comparative Economics* 39 (1): 65–81.
- Chinzara, Zivanemoyo, Sebastien Dessus, and Stephan Dreyhaupt. 2023. "How Institutional Reforms Can Attract More Private Investment." *INFRASTRUCTURE IN AFRICA*.
- Dorosh, Paul, Hyoung Gun Wang, Liangzhi You, and Emily Schmidt. 2012. "Road Connectivity, Population, and Crop Production in Sub-Saharan Africa." *Agricultural Economics* 43 (1): 89–103.
- Espinet, Xavier, Julie Rozenberg, Kulwinder Singh Rao, and Satoshi Ogita. 2018. "Piloting the Use of Network Analysis and Decision-Making under Uncertainty in Transport Operations: Preparation and Appraisal of a Rural Roads Project in Mozambique under Changing Flood Risk and Other Deep Uncertainties," June.
- Gupta, Sanjeev, Alvar Kangur, Chris Papageorgiou, and Abdoul Wane. 2014. "Efficiency-Adjusted Public Capital and Growth." *World Development* 57 (May): 164–78.
- limi, Atsushi. 2021. "Agricultural Production and Transport Connectivity: Evidence from Mozambique."

- Kilic Celik, S., M. A. Kose, F. Ohnsorge, and F. U. Ruch. 2023. "Potential Growth: A Global Database." Policy Research Working Paper 10354, World Bank, Washington, DC.
- Kremer, Stephanie, Alexander Bick, and Dieter Nautz. 2013. "Inflation and Growth: New Evidence from a Dynamic Panel Threshold Analysis." *Empirical Economics* 44 (2): 861–78.
- Malah Kuete, Yselle F., and Simplicie A. Asongu. 2023. "Infrastructure Development as a Prerequisite for Structural Change in Africa." *Journal of the Knowledge Economy* 14: 1386--1412.
- Matvejevs, Olegs, and Olegs Tkacevs. 2022. "Public Investment Crowds in Private Investment - with Ifs and Buts." *Working Papers, Working Papers*, , October.
- Mikou, Mehdi, Julie Rozenberg, Elco Koks, Charles Fox, and Tatiana Peralta Quiros. 2019. "Assessing Rural Accessibility and Rural Roads Investment Needs Using Open Source Data."
- Parry, Ian, Mr Simon Black, and Nate Vernon. 2021. *Still Not Getting Energy Prices Right: A Global and Country Update of Fossil Fuel Subsidies*. International Monetary Fund.
- Tunde, A. M., and E. E. Adeniyi. 2012. "Impact of Road Transport on Agricultural Development: A Nigerian Example." *Ethiopian Journal of Environmental Studies and Management* 5 (3): 232–38.
- Wiegand, Martin, Eric Koomen, Menno Pradhan, and Christopher Edmonds. 2023. "The Impact of Road Development on Household Welfare in Rural Papua New Guinea." *The Journal of Development Studies* 59 (6): 933–53.